

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short
portfolio, net of 10 bp transaction cost.
271 factors over 132 months, ordered by in-sample CAGR.

HOW TO READ

Each curve starts at 0 % in July 2015 and
compounds monthly long–short returns. The line is
coloured by the factor's style; the figure at the
right edge is the period-end cumulative return.

FACTOR STYLES

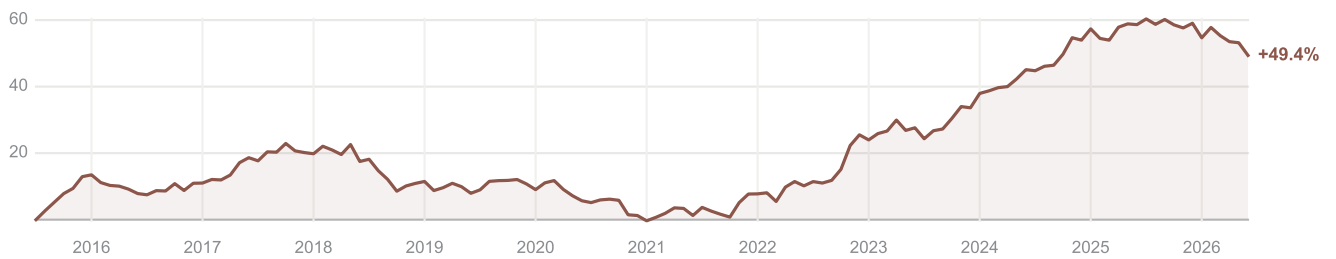
● Valuation	41	● Price Momentum	56
● Earnings Quality	55	● Size	1
● Analyst Expectations	24	● Historical Growth	56
● Capital Efficiency	28	● Volatility	10

001 In current model portfolio (50 factors)

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

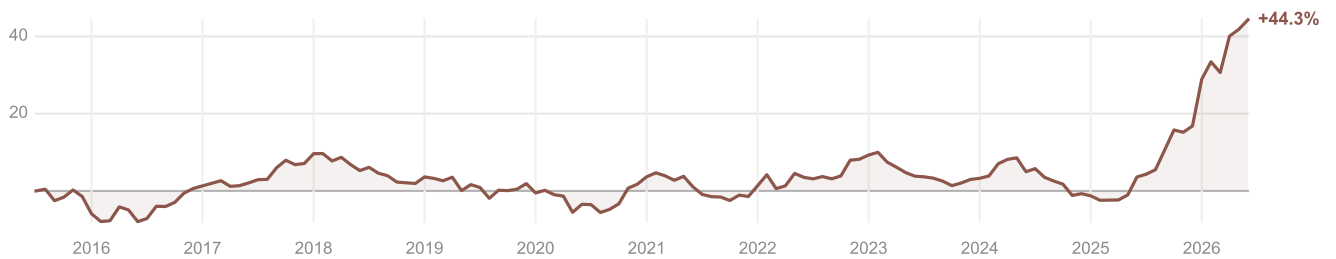
Historical Growth

CAGR +17.2%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

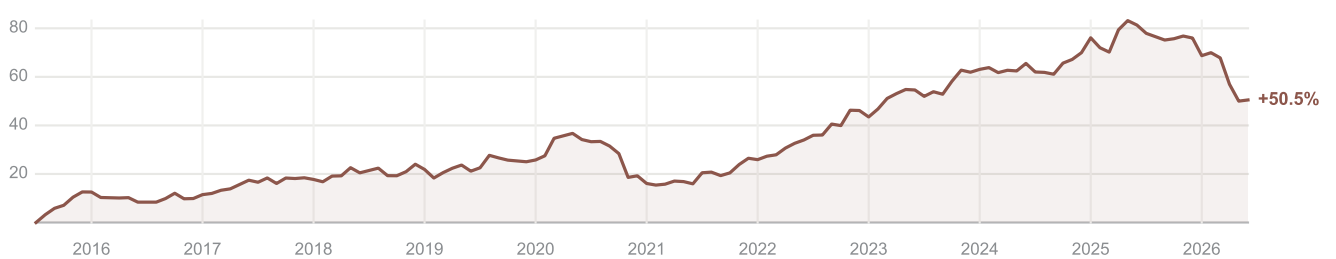
Historical Growth

CAGR +15.4%

**003 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

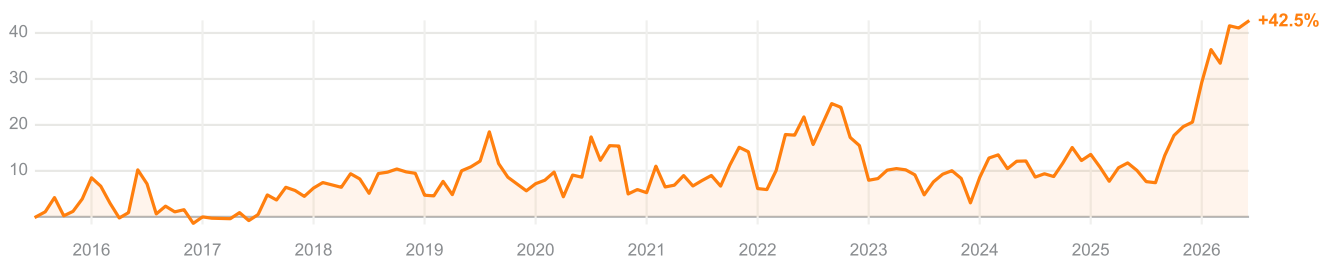
Historical Growth

CAGR +12.3%

**004 ● 50 Day to 200 Day Stock Price**

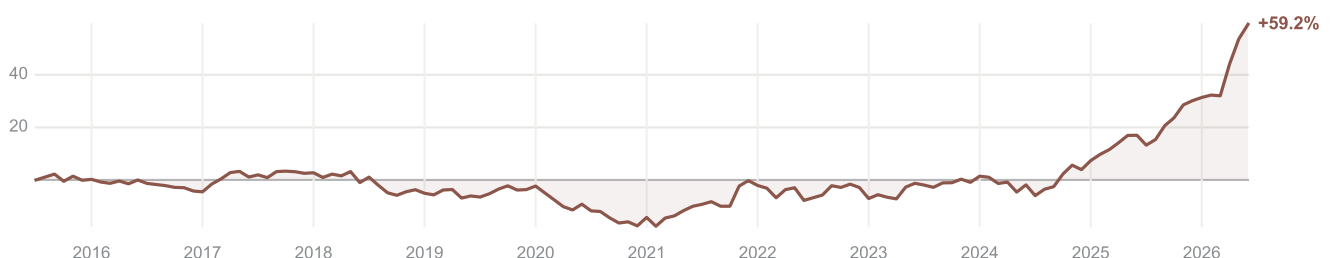
Price Momentum

CAGR +12.2%

**005 ● Change of Asset Turnover**

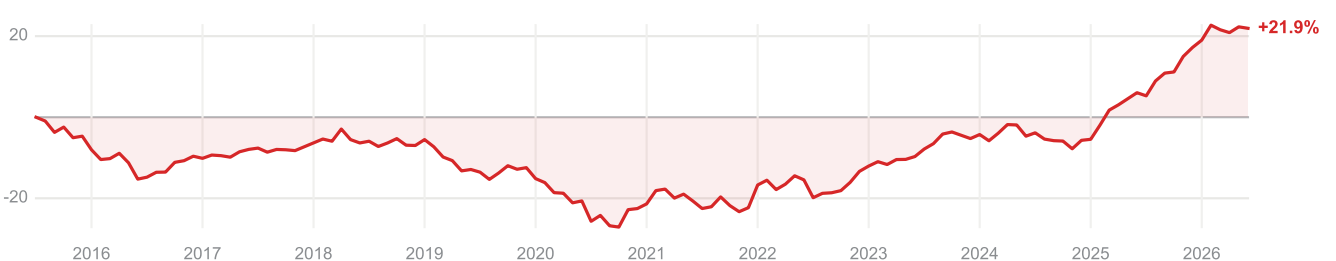
Historical Growth

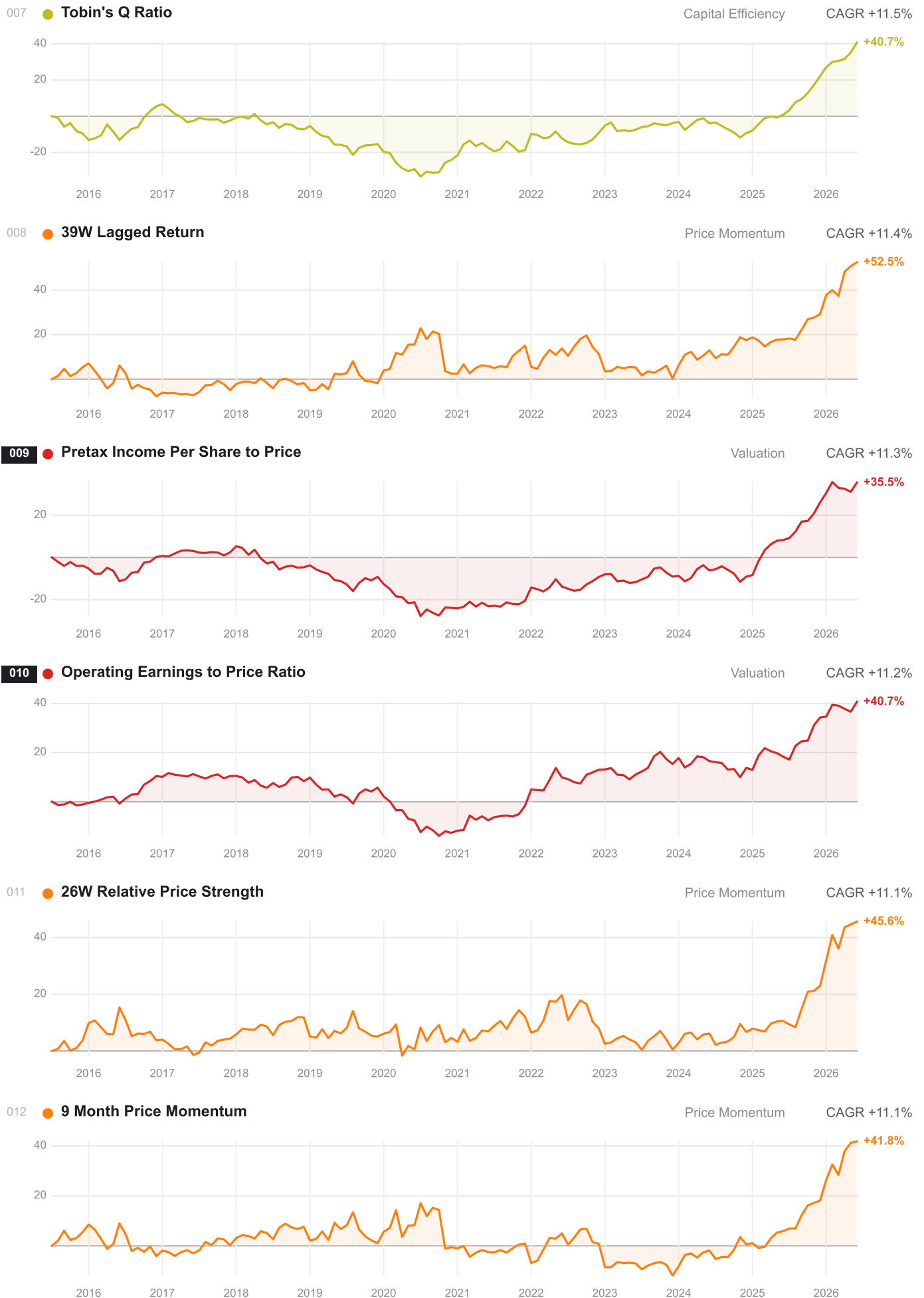
CAGR +11.8%

**006 ● Current Liability to Price**

Valuation

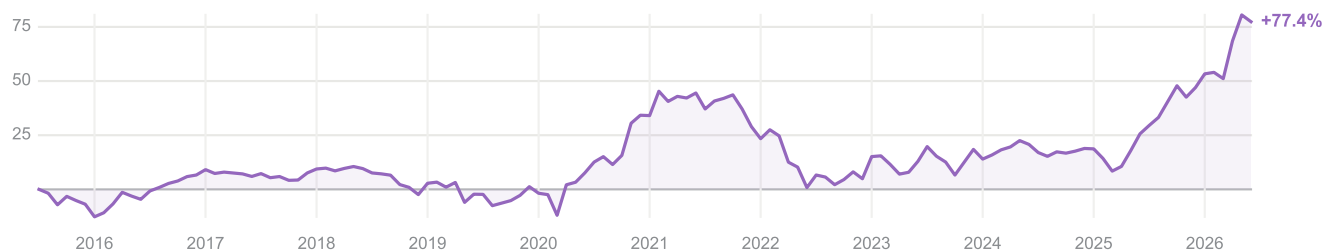
CAGR +11.7%



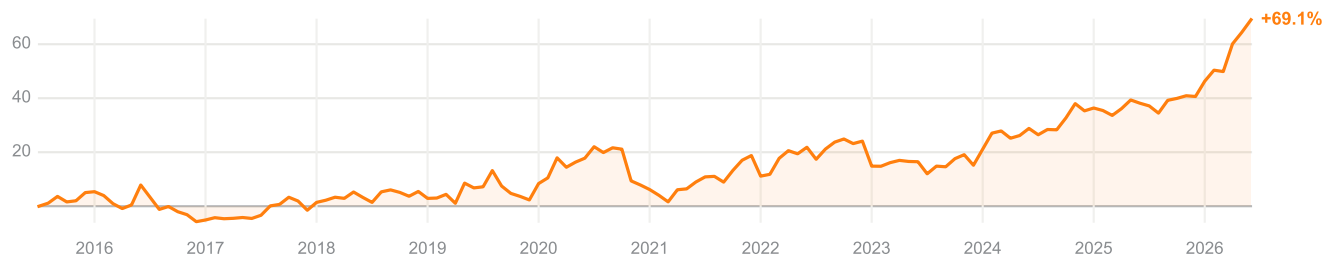


013 ● **12M Realized Price Volatility**

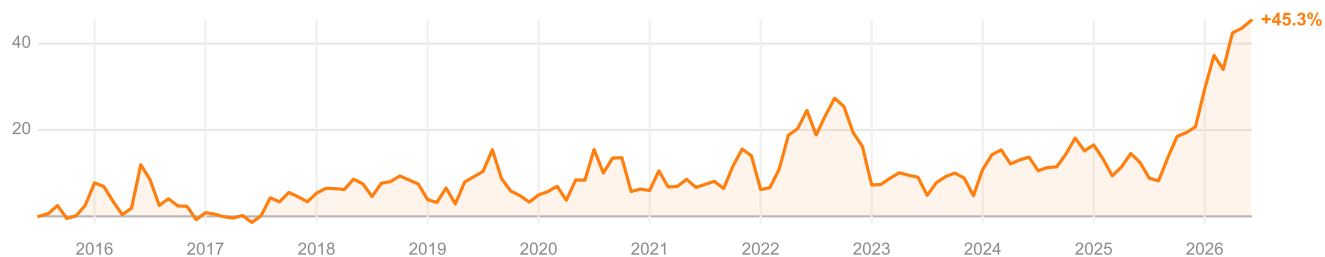
Volatility CAGR +11.0%

014 ● **Slope of 52W Price Trend**

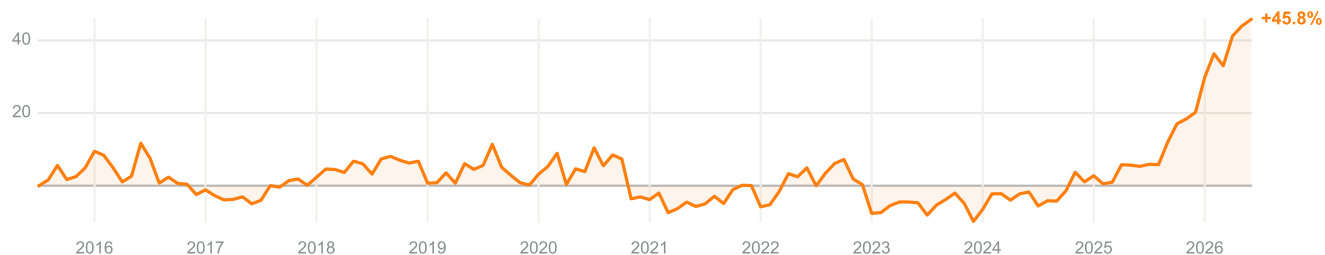
Price Momentum CAGR +11.0%

015 ● **15/36 Week Price Ratio**

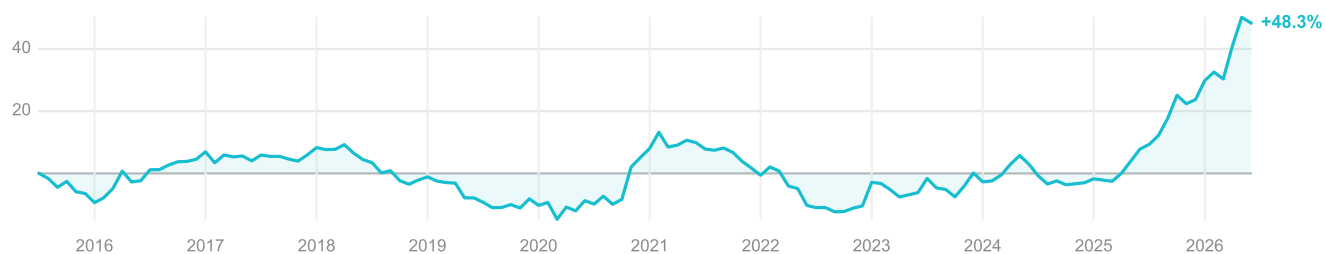
Price Momentum CAGR +10.9%

016 ● **4W to 52W Price Oscillator**

Price Momentum CAGR +10.8%

017 ● **Analyst Dispersion for FY2 EPS**

Analyst Expectations CAGR +10.7%

018 ● **Book to Price**

Valuation CAGR +10.6%



019 ● Normalized EP

Valuation CAGR +10.4%



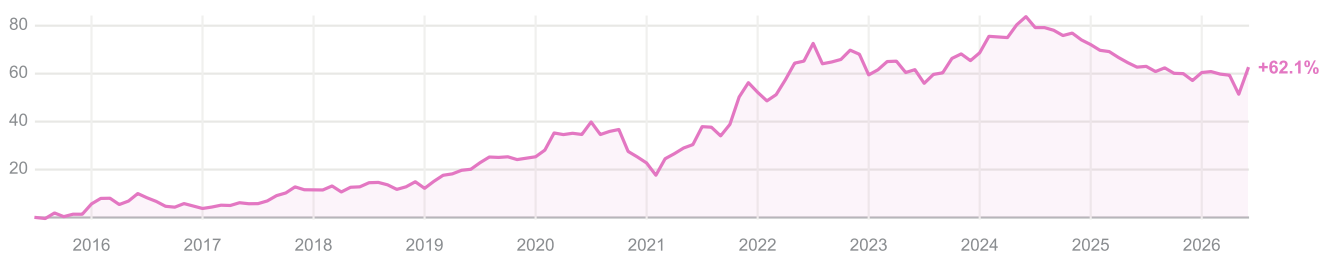
020 ● Indicated Annual Dividends to Price Ratio

Valuation CAGR +10.2%



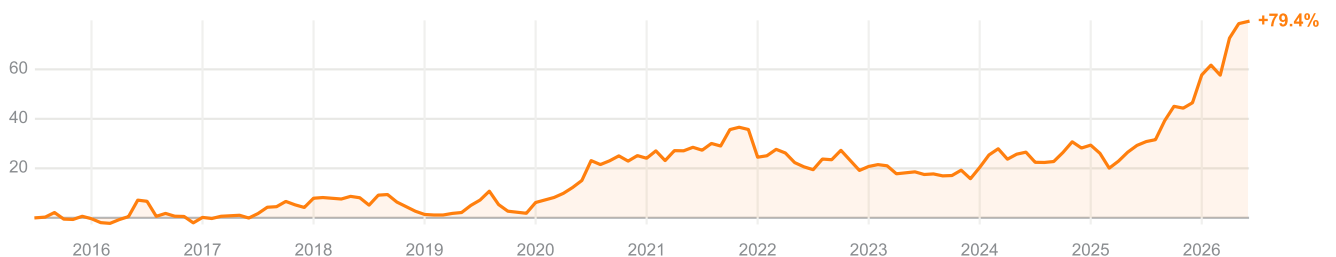
021 ● Operating Earnings to Assets Ratio

Earnings Quality CAGR +10.2%



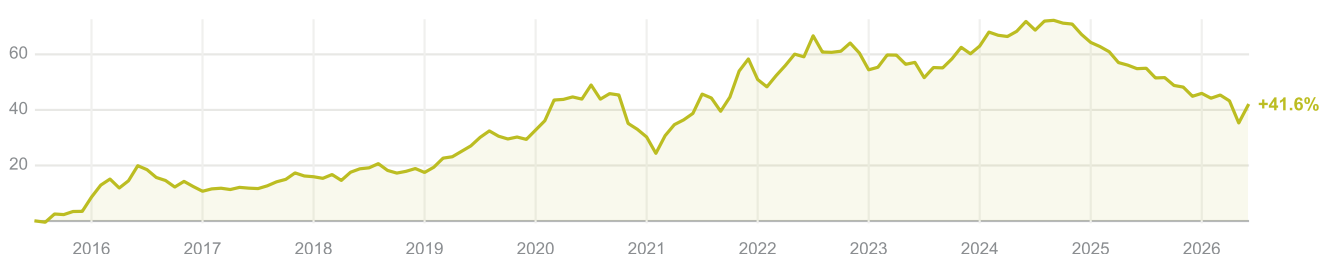
022 ● Percent Above 260-Day Low

Price Momentum CAGR +10.2%



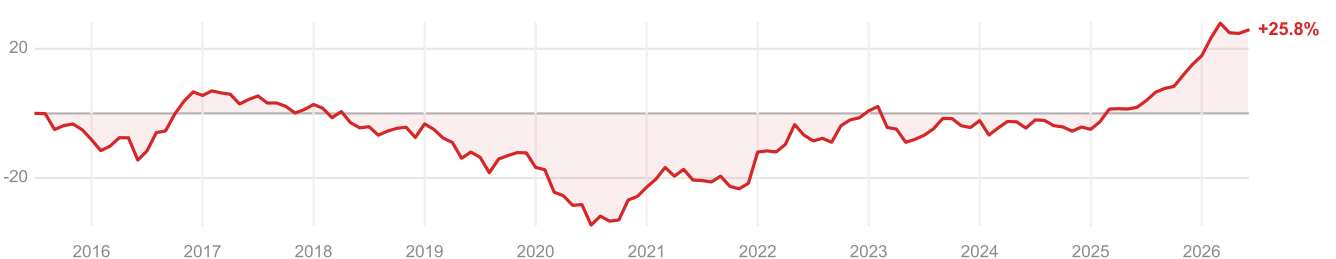
023 ● Return on Assets

Capital Efficiency CAGR +10.1%



024 ● EBITDA to Price

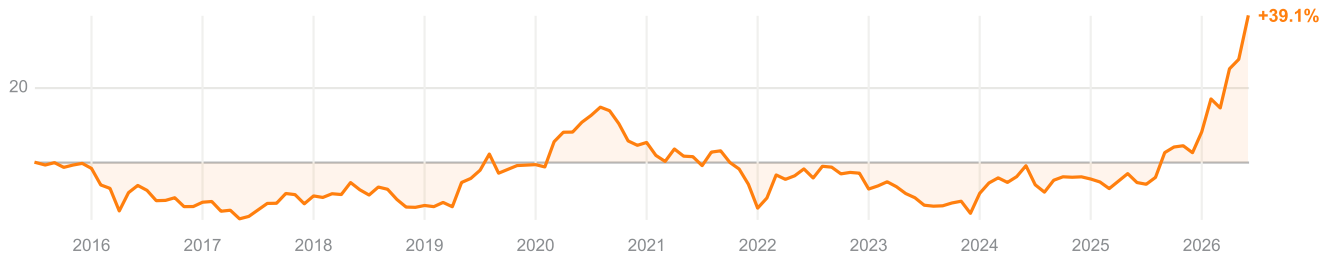
Valuation CAGR +10.0%



025 ● **Stability of Change in Number of Holders**

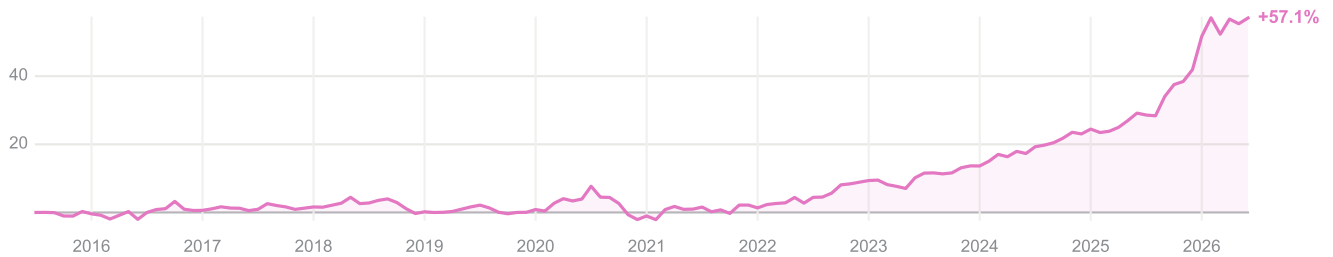
Price Momentum

CAGR +10.0%

026 ● **Unexpected Inventory Change**

Earnings Quality

CAGR +9.9%

027 ● **Log of Total Last Quarter Assets**

Size

CAGR +9.8%

028 ● **Assets to Price Ratio**

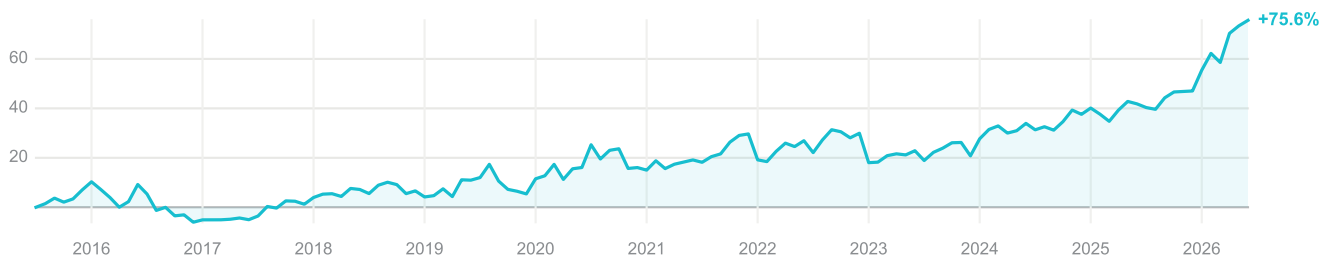
Valuation

CAGR +9.8%

029 ● **6M Chg in Target Price**

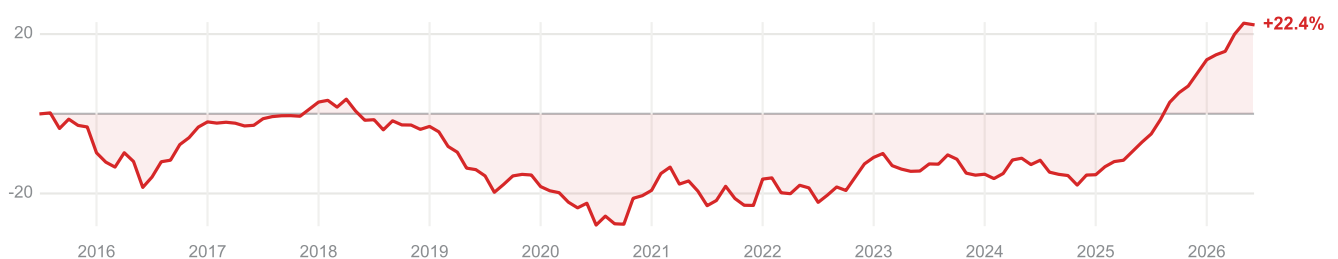
Analyst Expectations

CAGR +9.7%

030 ● **Cash to Price**

Valuation

CAGR +9.7%



031 ● Sales to Price Ratio

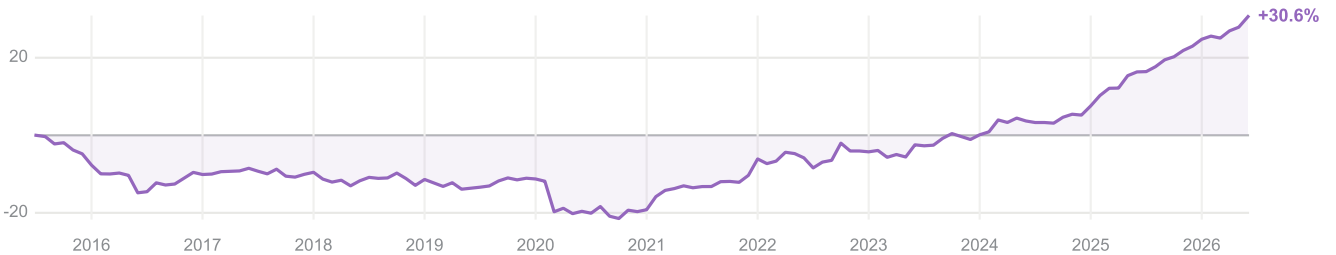
Valuation CAGR +9.7%

**032 ● Growth Flow to Price**

Valuation CAGR +9.6%

**033 ● Book Leverage**

Volatility CAGR +9.6%

**034 ● Earnings to Price**

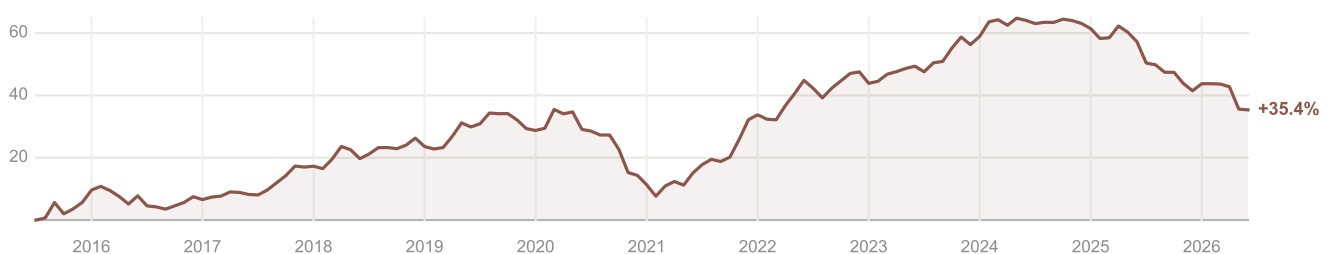
Valuation CAGR +9.4%

**035 ● Cash Flow to Total Asset**

Earnings Quality CAGR +9.4%

**036 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings**

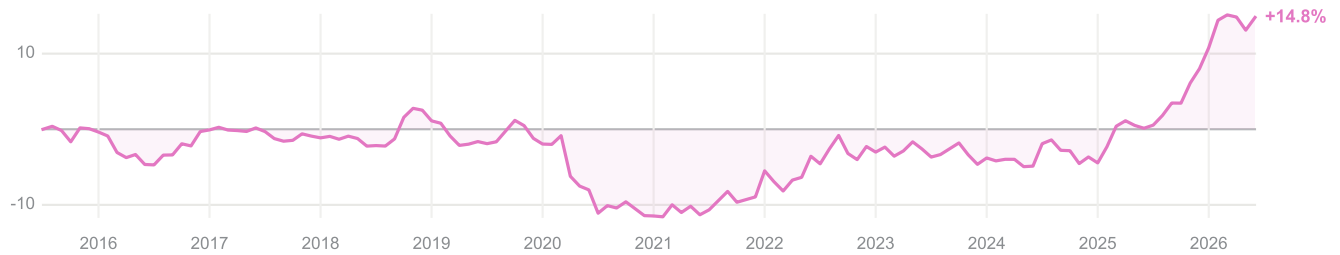
Historical Growth CAGR +9.2%



037 ● ROA 20 Qtr Standard Deviation

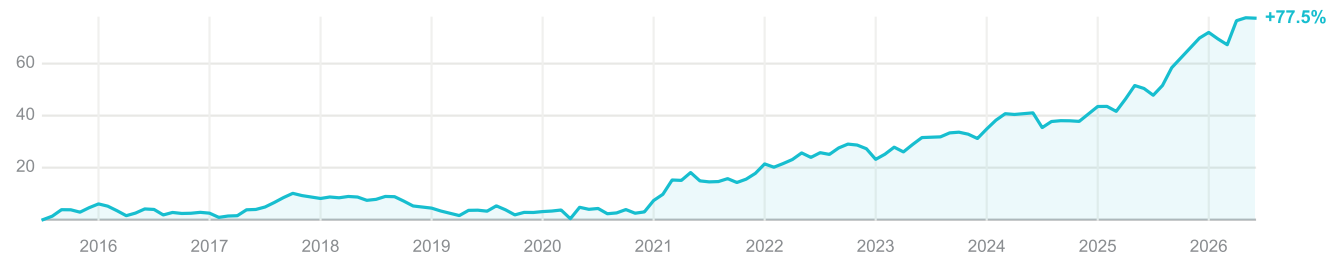
Earnings Quality

CAGR +9.2%

**038 ● Adjusted 3M Revision in FY1 EPS Estimate**

Analyst Expectations

CAGR +9.2%

**039 ● Inverse P/E Ratio Adj for Growth and Yield**

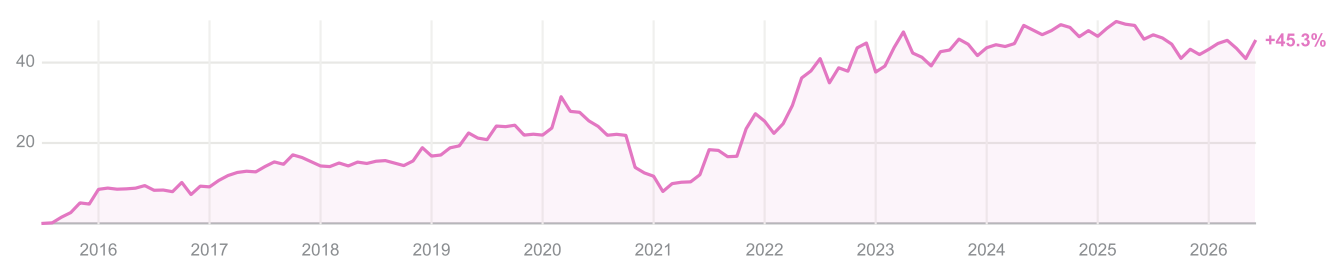
Valuation

CAGR +9.1%

**040 ● Trailing 12-Month Earnings/Trailing 12-Month Sales**

Earnings Quality

CAGR +9.1%

**041 ● Net Cash Flow to Enterprise Value**

Valuation

CAGR +9.1%

**042 ● Solvency Ratio**

Valuation

CAGR +9.1%



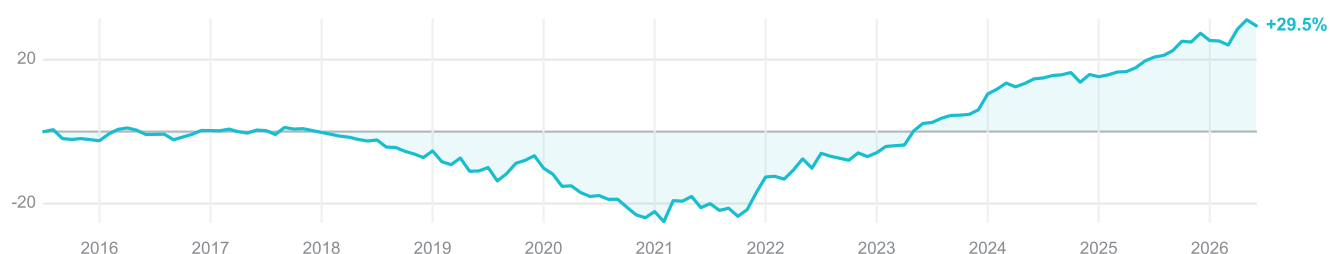
043 ● Dividends to Price Ratio

Valuation CAGR +9.1%



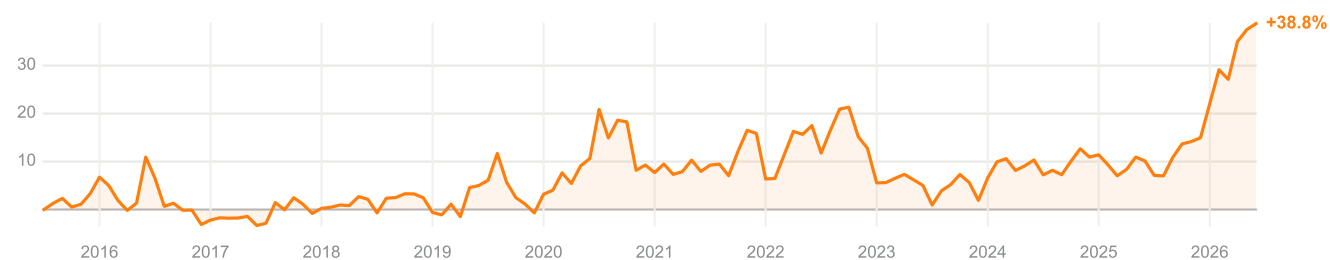
044 ● Adjusted Inverse PEG

Analyst Expectations CAGR +9.1%



045 ● 180D Price TStat

Price Momentum CAGR +8.9%



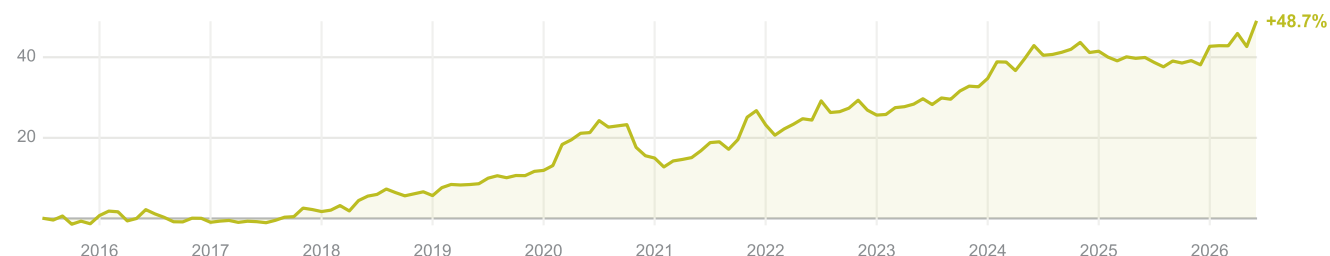
046 ● Adjusted EBIT to Price

Valuation CAGR +8.9%



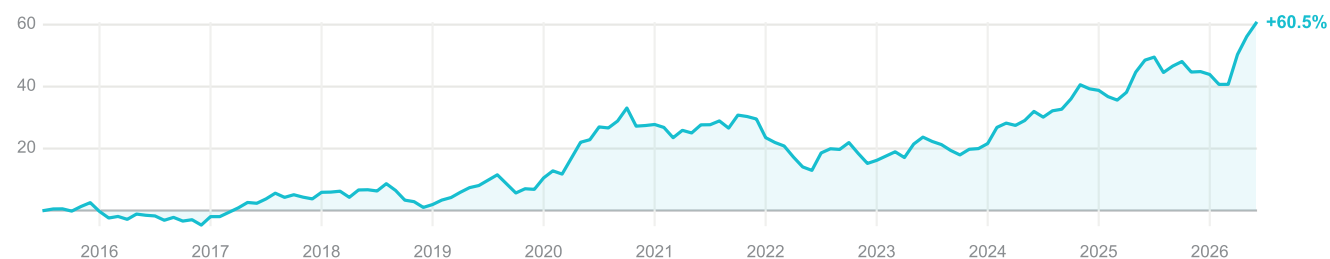
047 ● EBITDA to Assets

Capital Efficiency CAGR +8.8%



048 ● Expected LTG

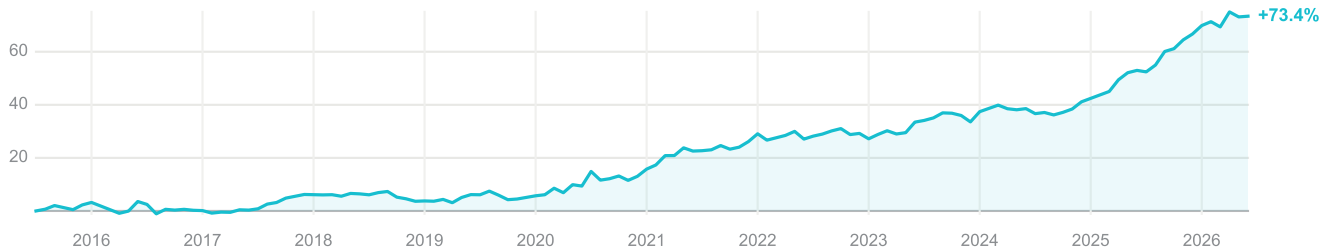
Analyst Expectations CAGR +8.7%



049 3M Revision in FY1 EPS Estimate

Analyst Expectations

CAGR +8.7%

**050 Adj 1Y Chg in Asset Adjusted FCF**

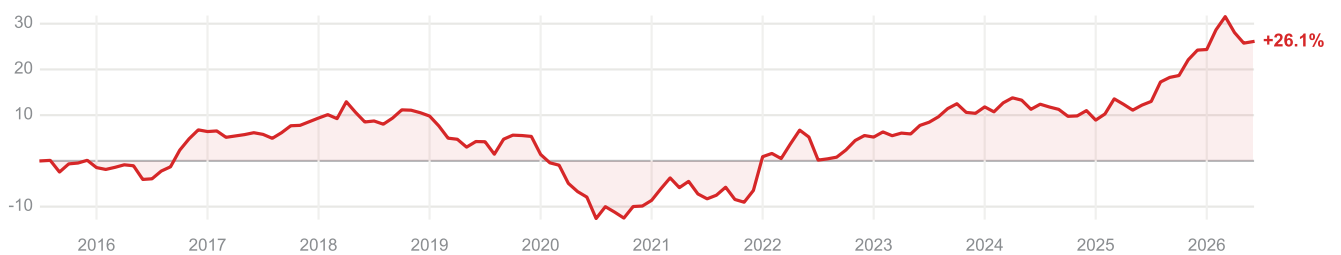
Historical Growth

CAGR +8.5%

**051 Sales to EV Ratio**

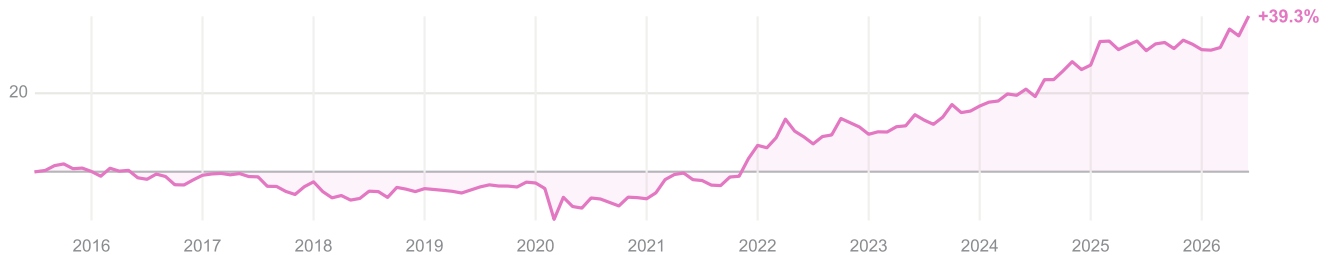
Valuation

CAGR +8.5%

**052 Working Capital to Sales**

Earnings Quality

CAGR +8.5%

**053 Adjusted Book to Enterprise Value**

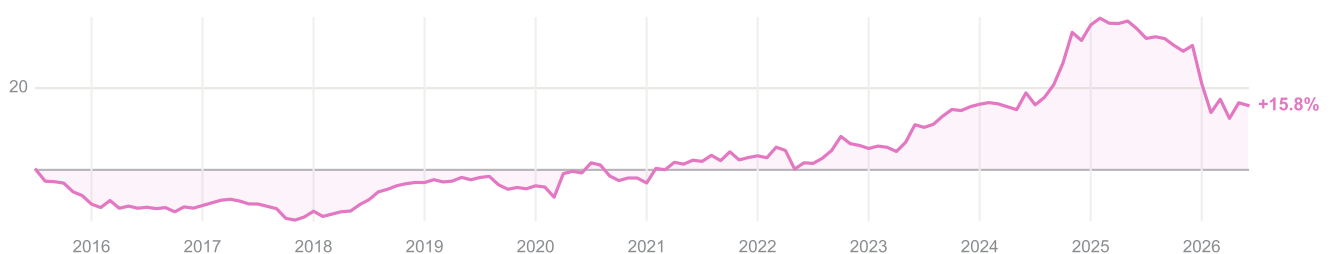
Valuation

CAGR +8.5%

**054 Balance Sheet Accruals**

Earnings Quality

CAGR +8.4%



055 ● EBITDA to Enterprise Value

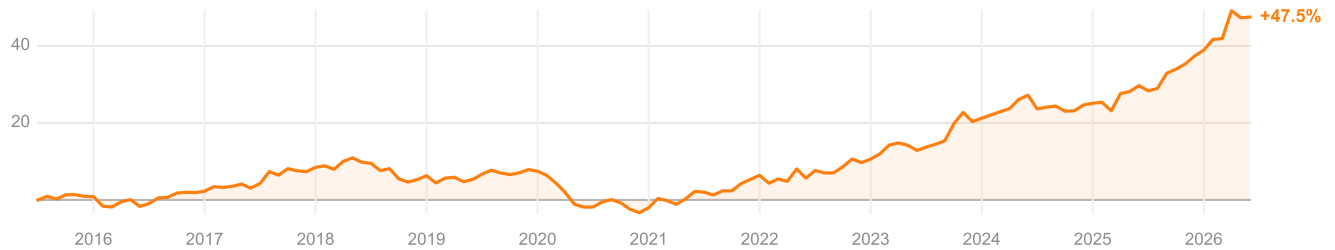
Valuation

CAGR +8.4%

**056 ● Short Interest Trading Volume**

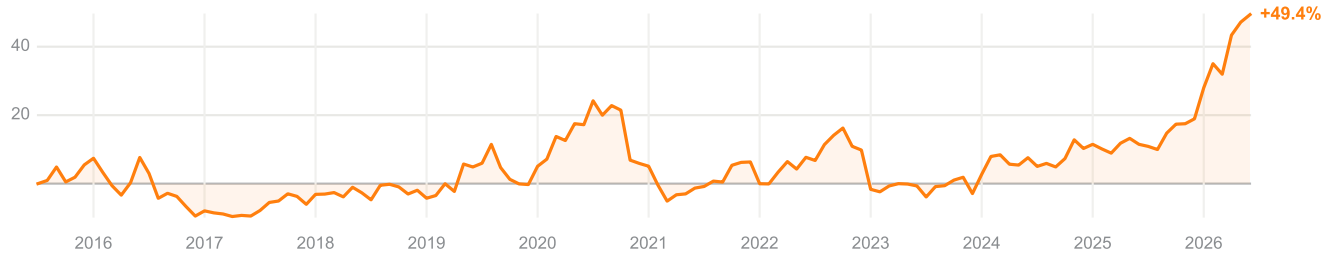
Price Momentum

CAGR +8.4%

**057 ● 12M - 1M Price Momentum**

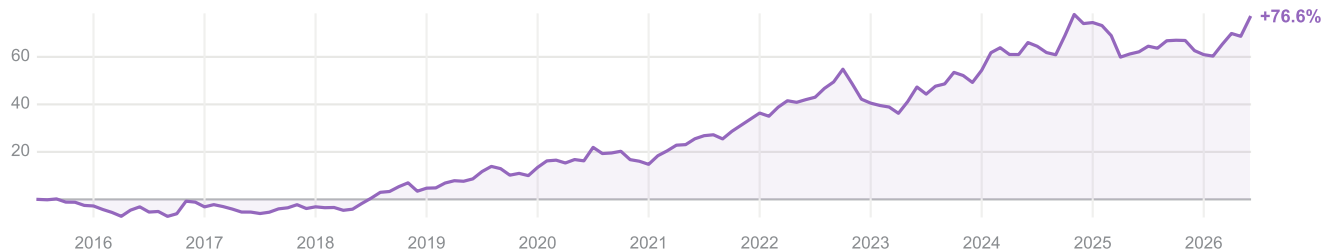
Price Momentum

CAGR +8.3%

**058 ● Amihud Illiquidity Measure**

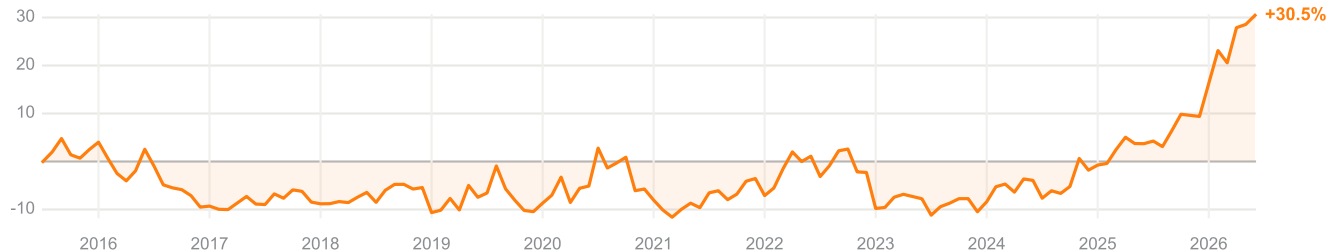
Volatility

CAGR +8.3%

**059 ● 52W High Low**

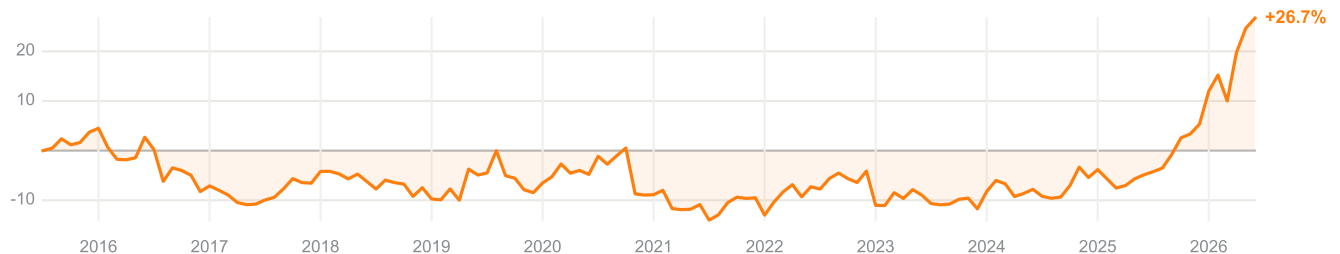
Price Momentum

CAGR +8.3%

**060 ● 52-Week Volume Price Trend with 20-Day Lag**

Price Momentum

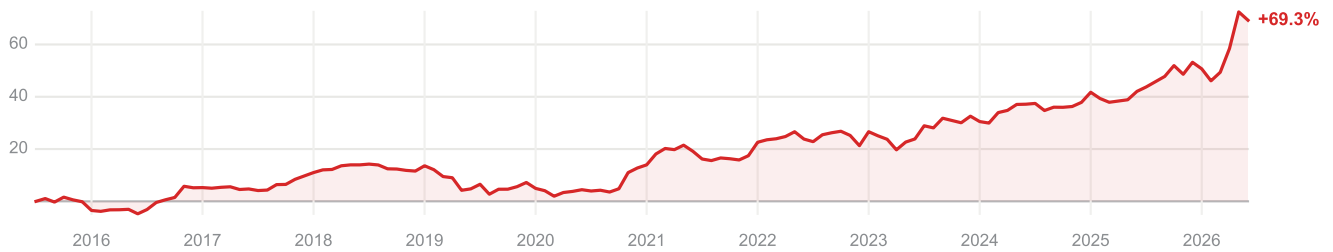
CAGR +8.2%



061 ● **Cash to Adjusted Enterprise Value**

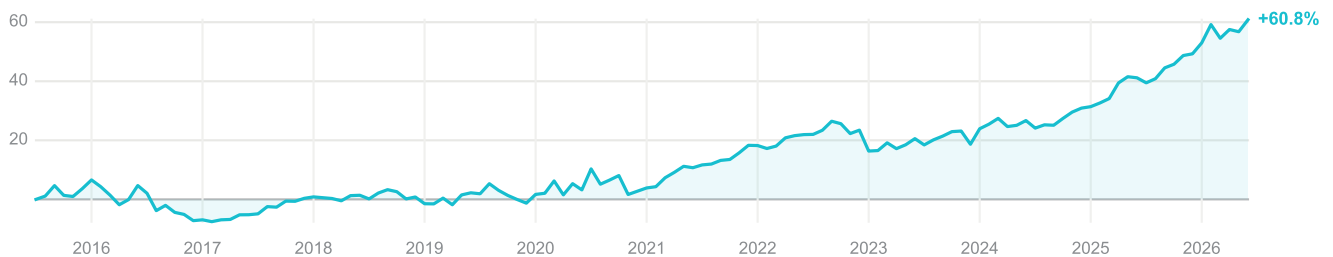
Valuation

CAGR +8.2%

062 ● **3M Revision in FY2 EPS Estimate**

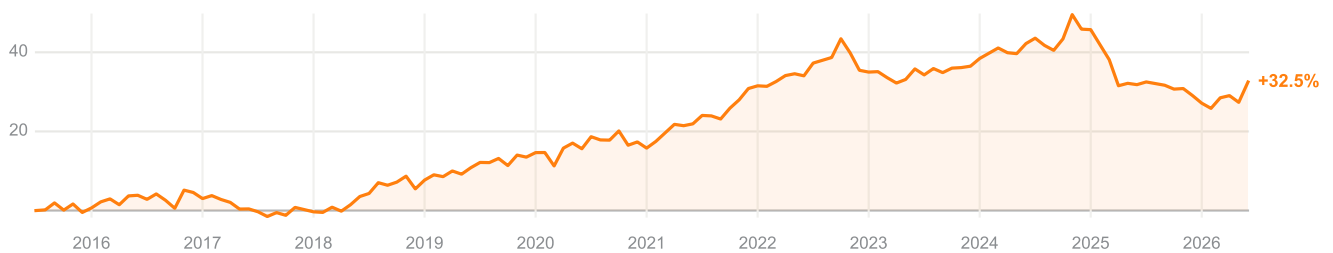
Analyst Expectations

CAGR +8.1%

063 ● **Institution Ownership Level - Active Manager**

Price Momentum

CAGR +8.1%

064 ● **Operating Cash Flow to Price**

Valuation

CAGR +8.1%

065 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

Historical Growth

CAGR +8.0%

066 ● **Log of Unadjusted Stock Price**

Price Momentum

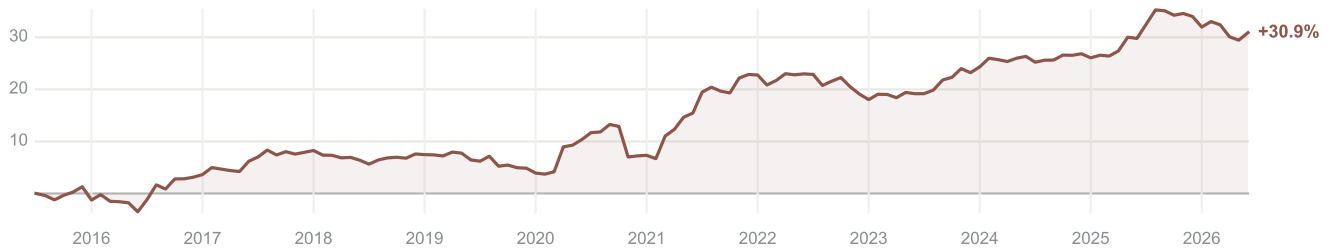
CAGR +8.0%



067 ● 3Y Chg in Operating Cash Flow to Price

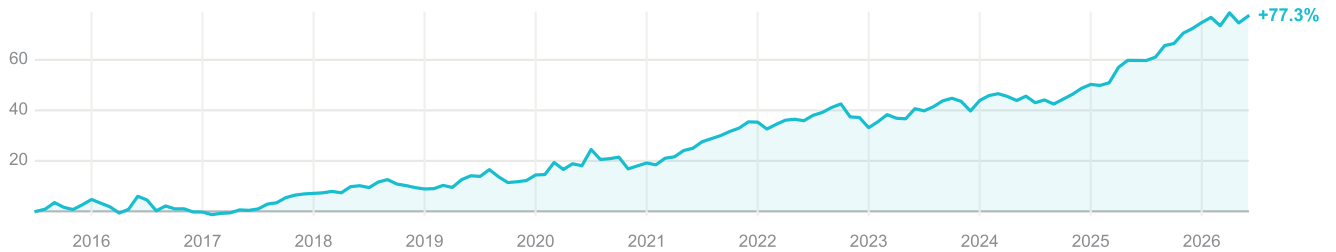
Historical Growth

CAGR +8.0%

**068 ● Street Revision Magnitude**

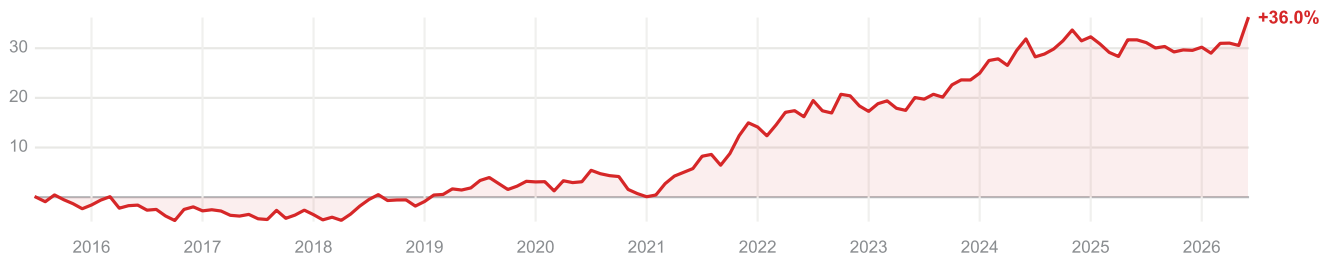
Analyst Expectations

CAGR +7.9%

**069 ● Cash Flow to Equity**

Valuation

CAGR +7.9%

**070 ● Operating Cash Flow to Invested Capital**

Earnings Quality

CAGR +7.9%

**071 ● Forward Earnings to Price**

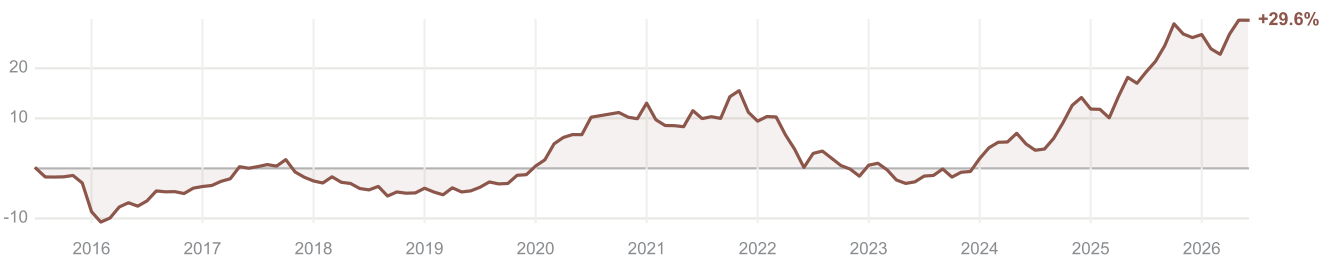
Valuation

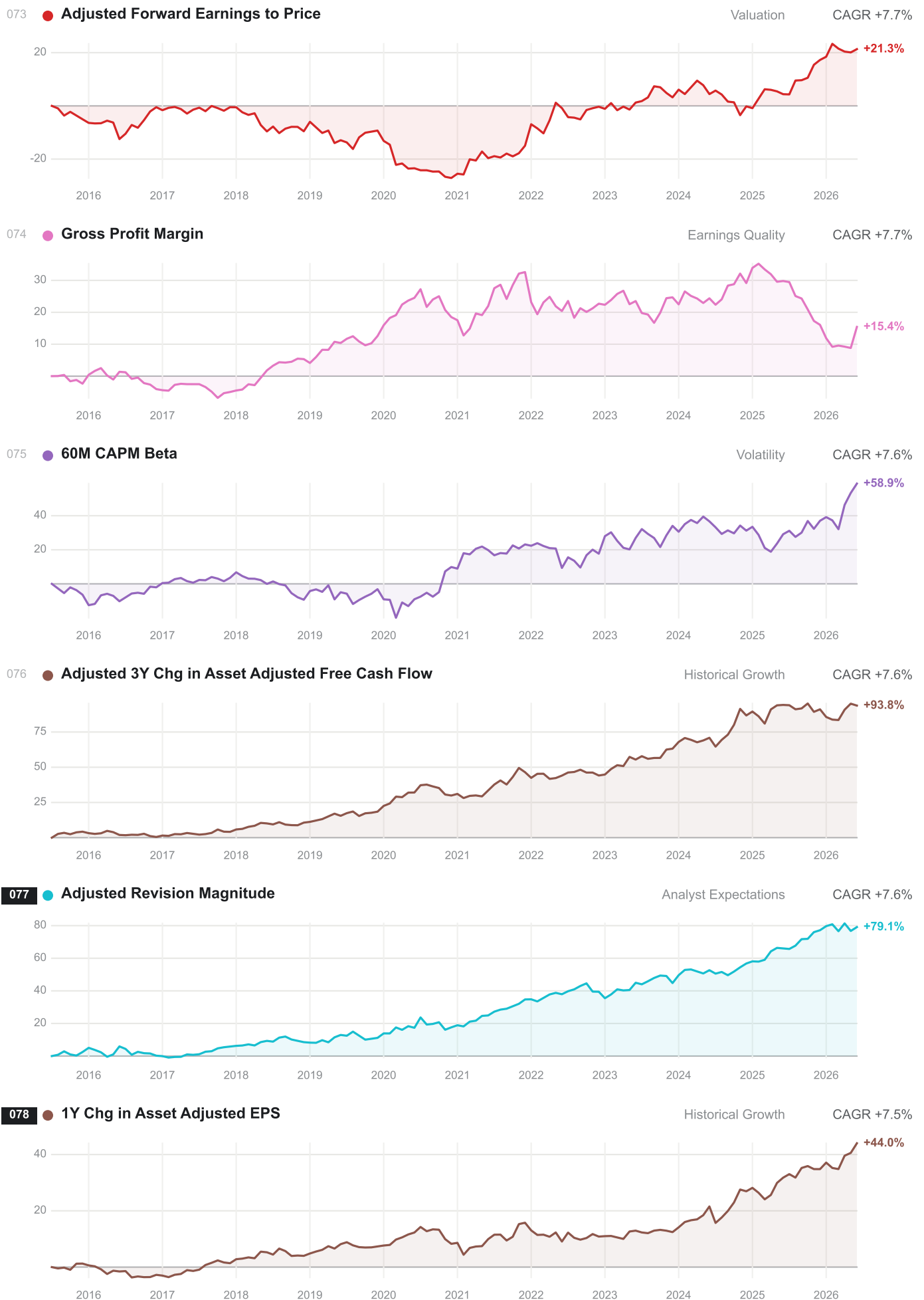
CAGR +7.9%

**072 ● 12-Quarter Trendline in Trailing 12-Month Earnings**

Historical Growth

CAGR +7.7%





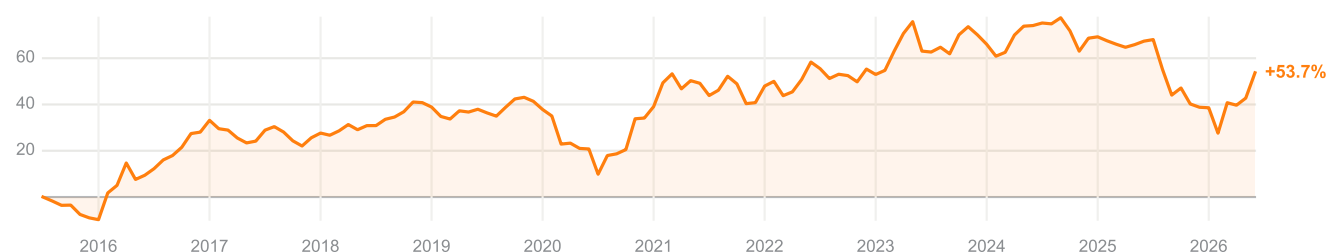
079 ● Net Cash Flow to Price

Valuation CAGR +7.5%



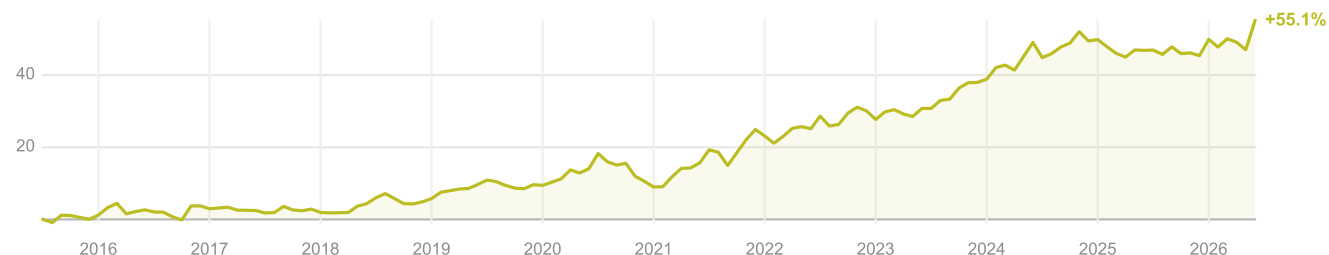
080 ● Sharpe Ratio

Price Momentum CAGR +7.3%



081 ● Return on Equity

Capital Efficiency CAGR +7.2%



082 ● Operating Cash Flow to Enterprise Value

Valuation CAGR +7.2%



083 ● 1Y Chg in Operating Margin

Historical Growth CAGR +7.2%



084 ● 1Y Chg in Free Cash Flow per Share

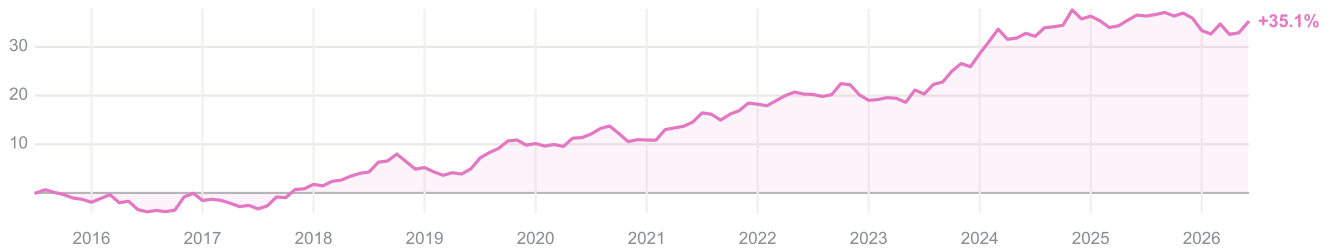
Historical Growth CAGR +7.1%



085 Equity Turnover

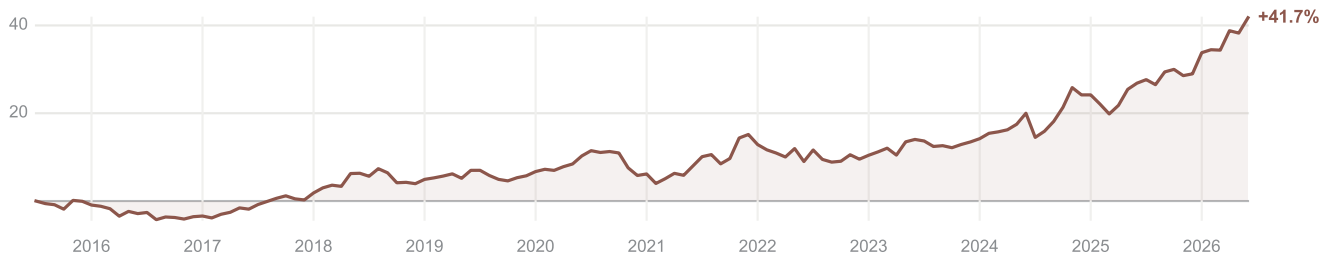
Earnings Quality

CAGR +7.1%

**086 1Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +7.1%

**087 90 Day Coefficient of Variation**

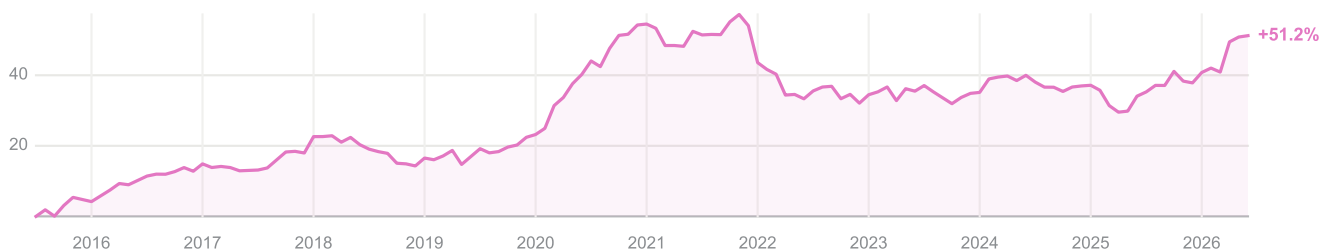
Volatility

CAGR +7.1%

**088 Working Capital to Assets Ratio**

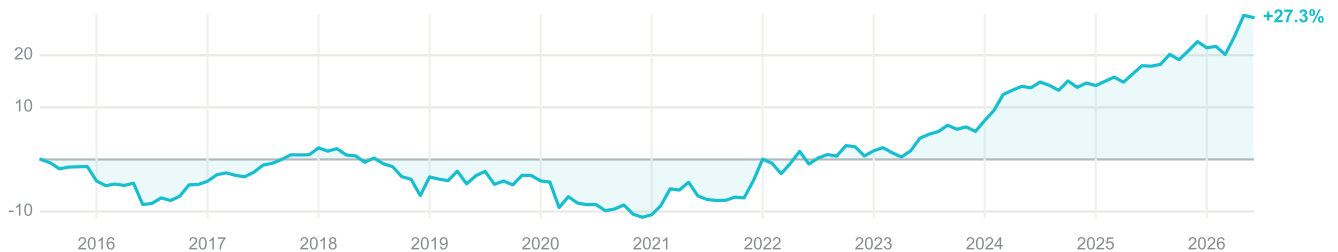
Earnings Quality

CAGR +7.0%

**089 Inverse PEG**

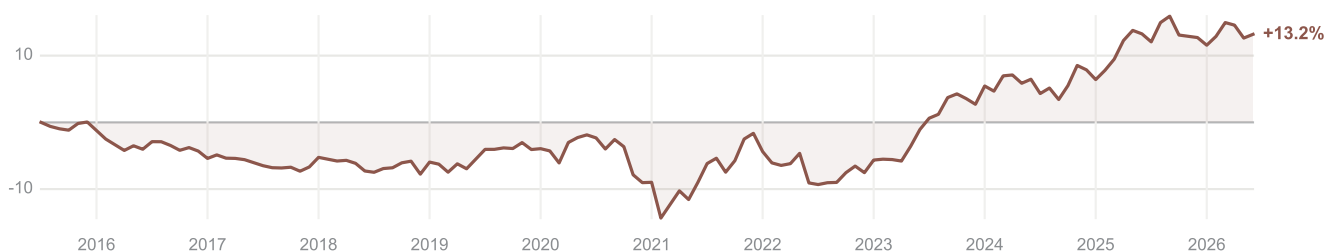
Analyst Expectations

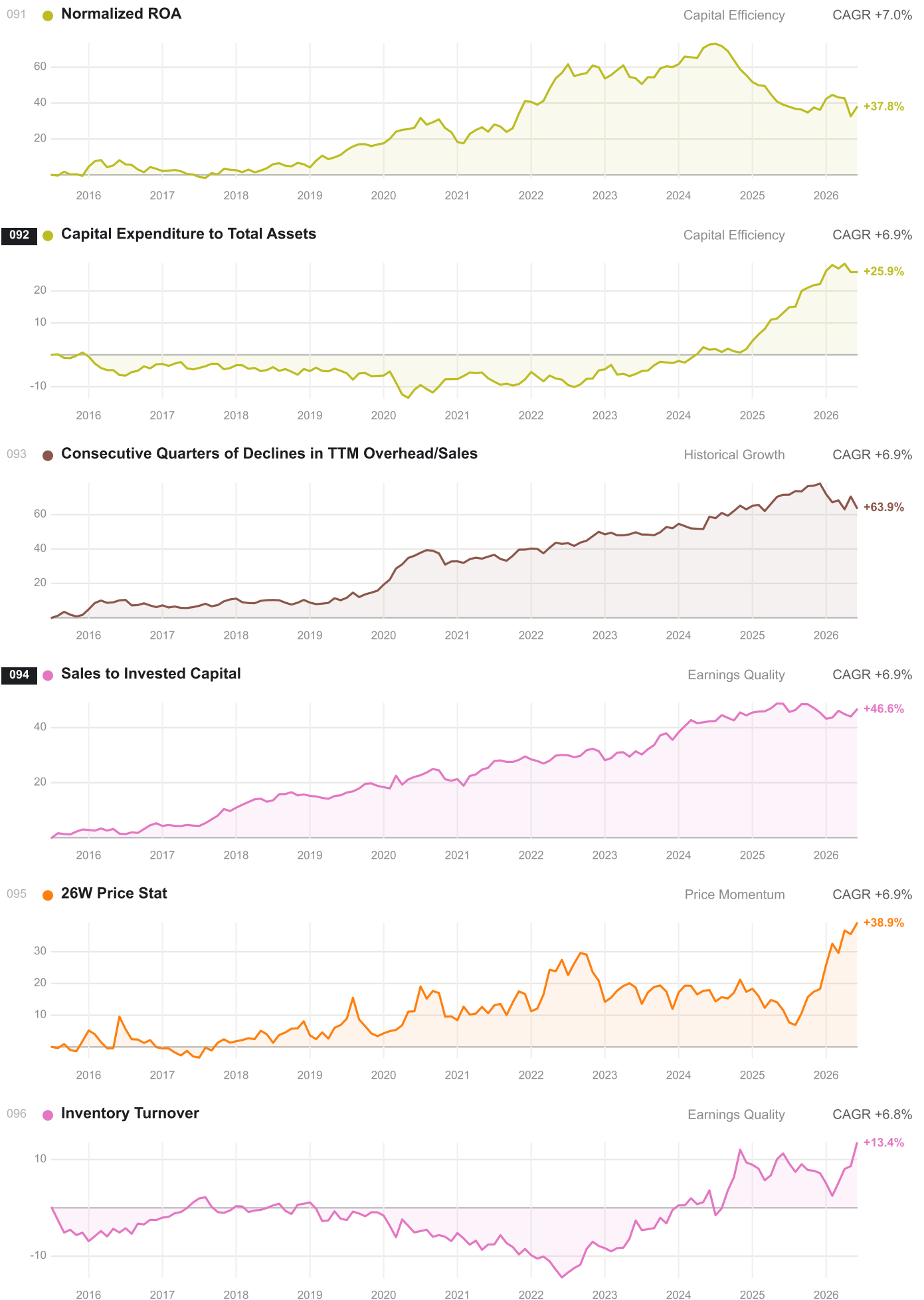
CAGR +7.0%

**090 1Y Chg in Sales to Price**

Historical Growth

CAGR +7.0%

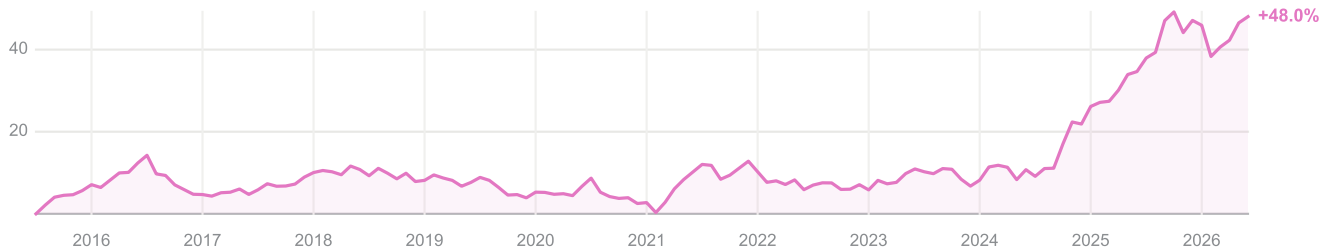




097 ● Year over Year Change in SGA to Sales

Earnings Quality

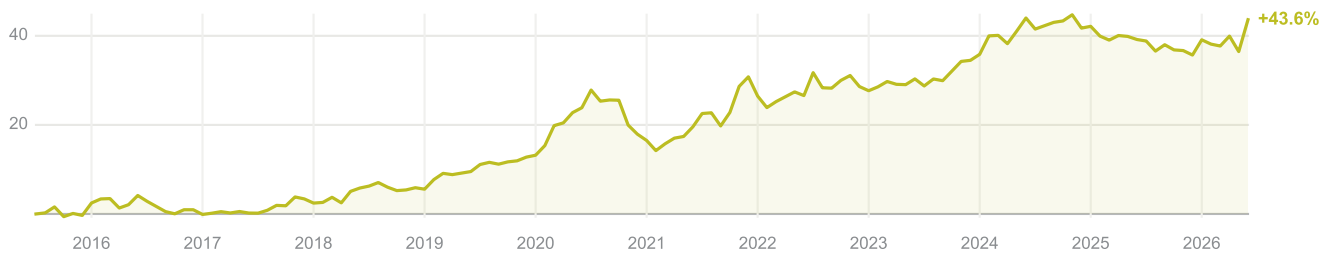
CAGR +6.7%



098 ● EBIT to Assets

Capital Efficiency

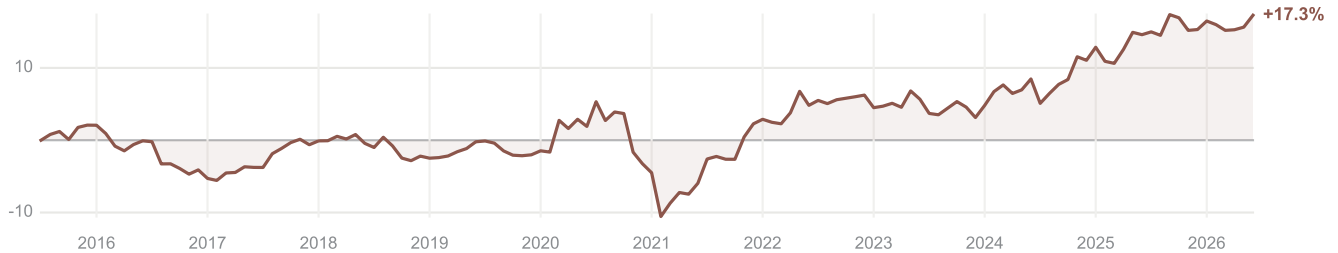
CAGR +6.7%



099 ● 1Y Chg in Earnings to Price

Historical Growth

CAGR +6.7%



100 ● Long Term Debt to Equity Ratio

Capital Efficiency

CAGR +6.7%



101 ● Adj. EBITDA to Enterprise Value

Valuation

CAGR +6.6%

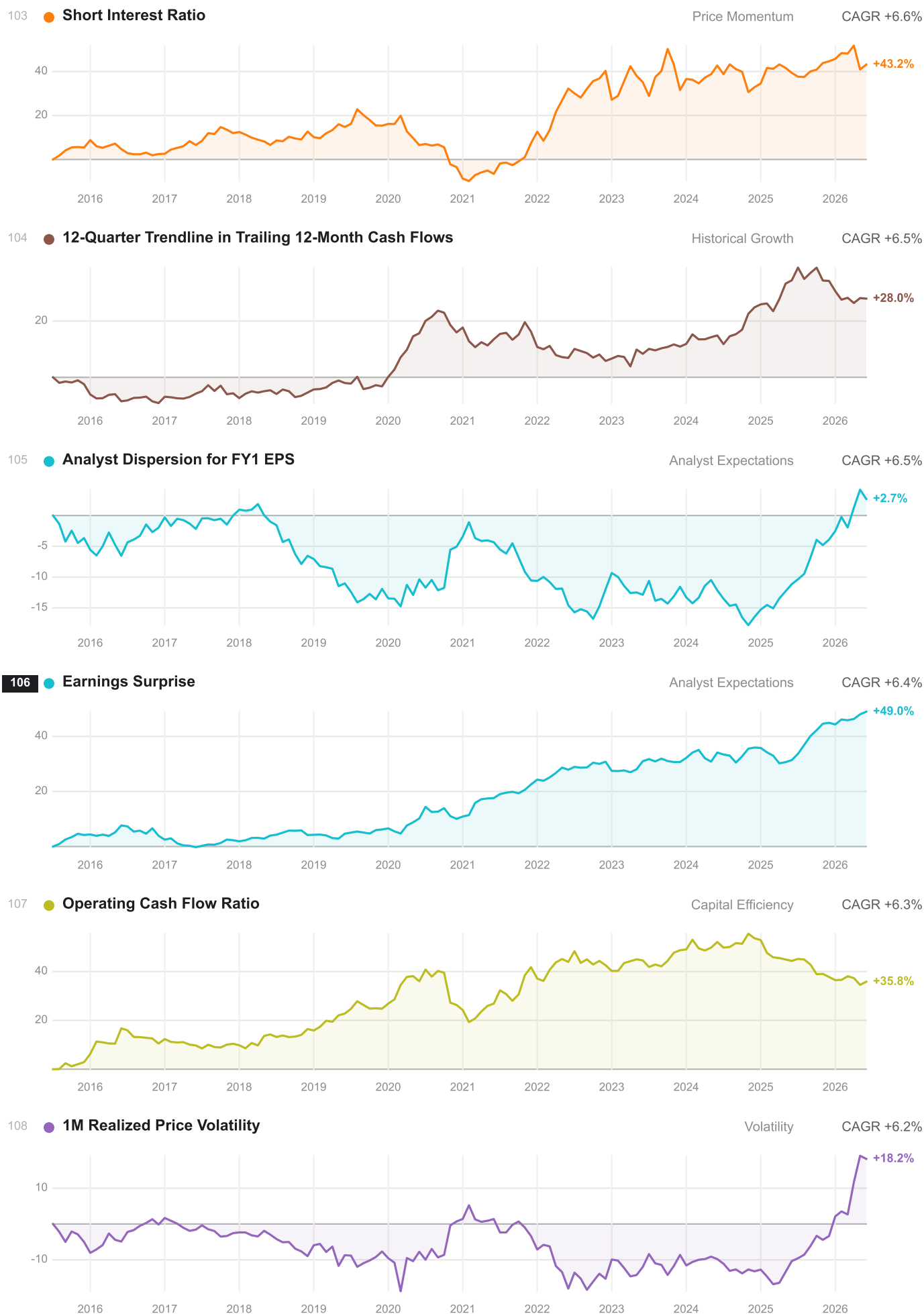


102 ● Adj 1Y Chg in Earnings per Share

Historical Growth

CAGR +6.6%

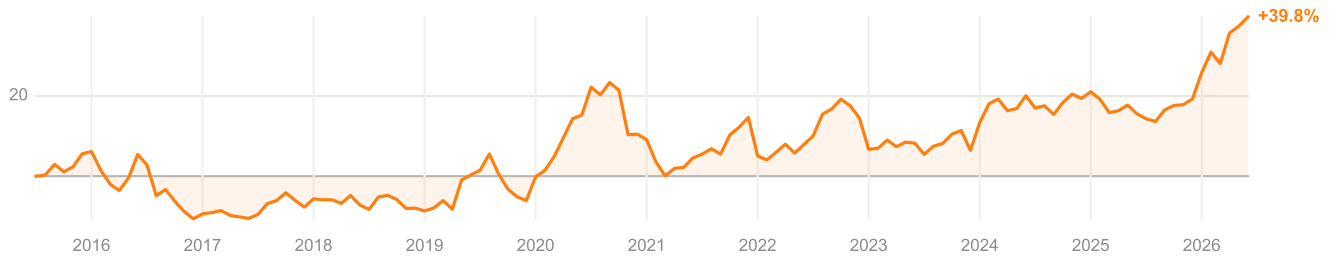




109 ● Volatility adjusted 12M return

Price Momentum

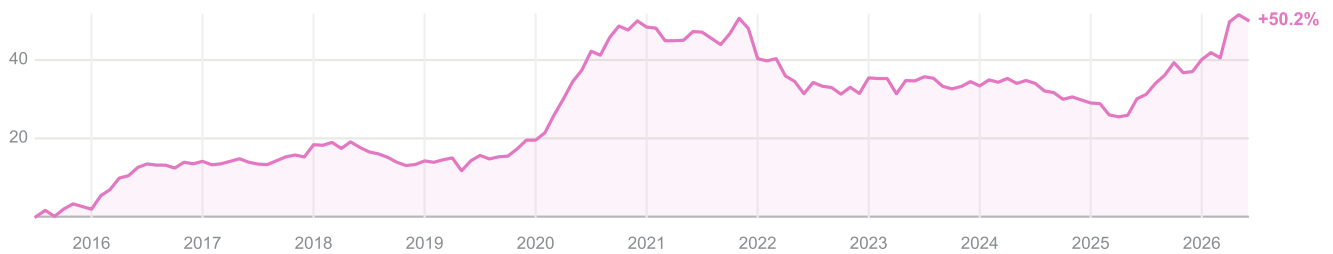
CAGR +6.2%



110 ● Current Ratio

Earnings Quality

CAGR +6.2%



111 ● Cash to EV Ratio

Valuation

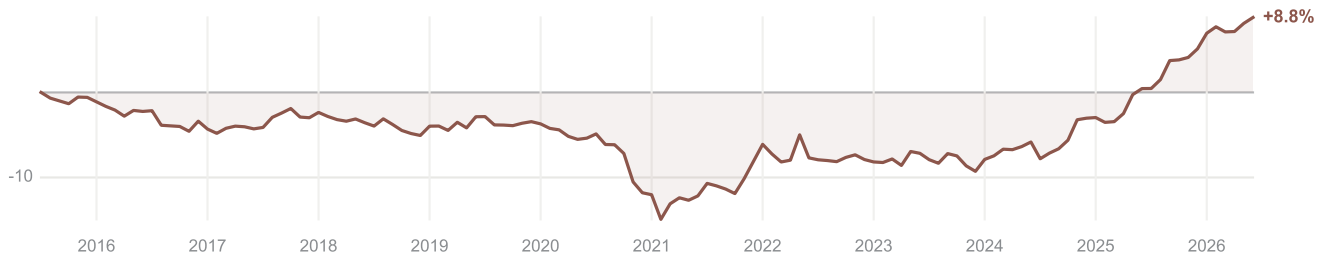
CAGR +6.2%



112 ● 1Y Chg in Cash Flow to Price

Historical Growth

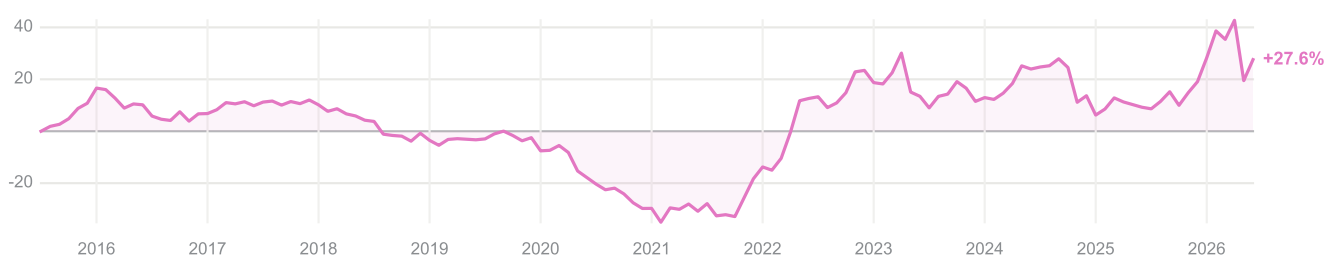
CAGR +6.1%



113 ● EBITDA Margin

Earnings Quality

CAGR +6.1%



114 ● Adj Free Cash Flow to Price

Valuation

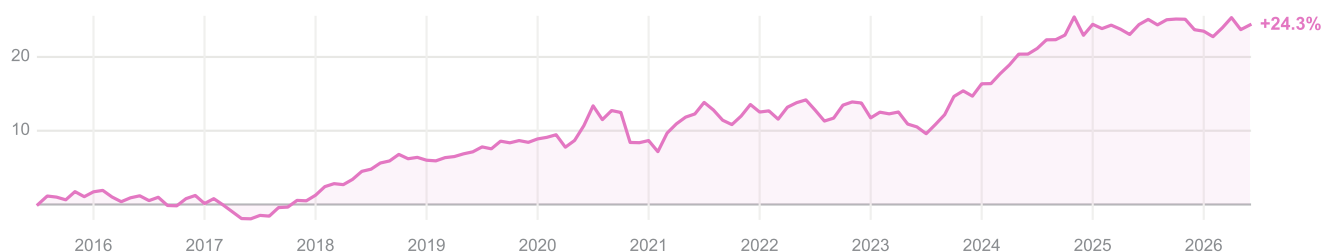
CAGR +6.0%



115 Capital Expenditure to Sales

Earnings Quality

CAGR +6.0%

**116 Trailing 12-Month Overhead/Trailing 12-Month Sales**

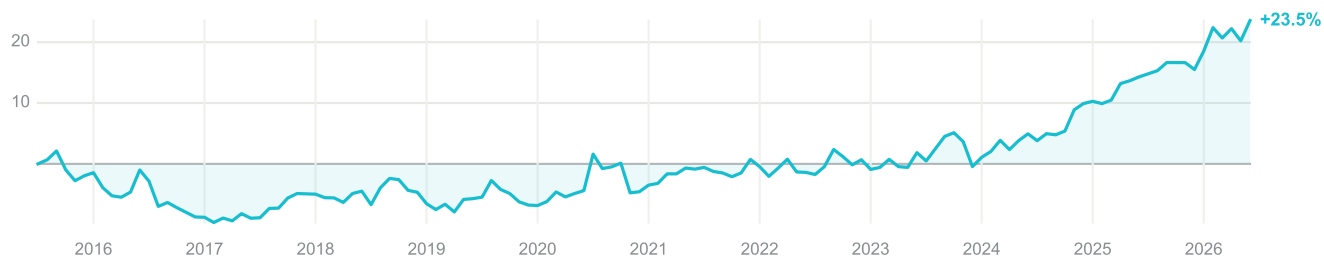
Historical Growth

CAGR +6.0%

**117 Adj Number of EPS FY2 Revisions**

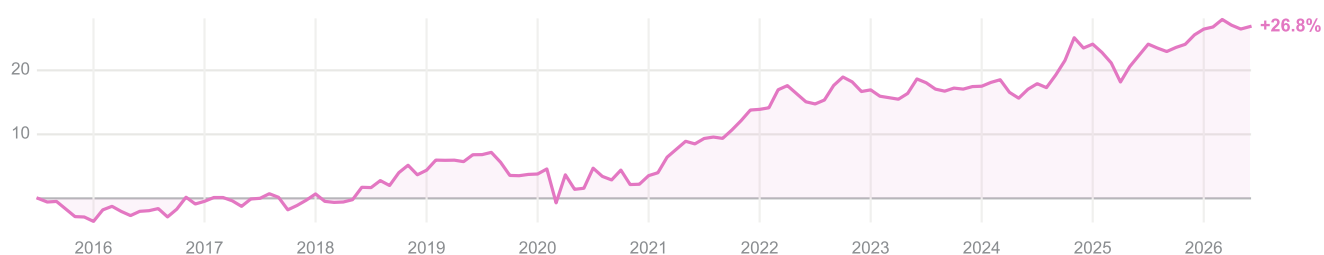
Analyst Expectations

CAGR +6.0%

**118 Working Capital Accruals**

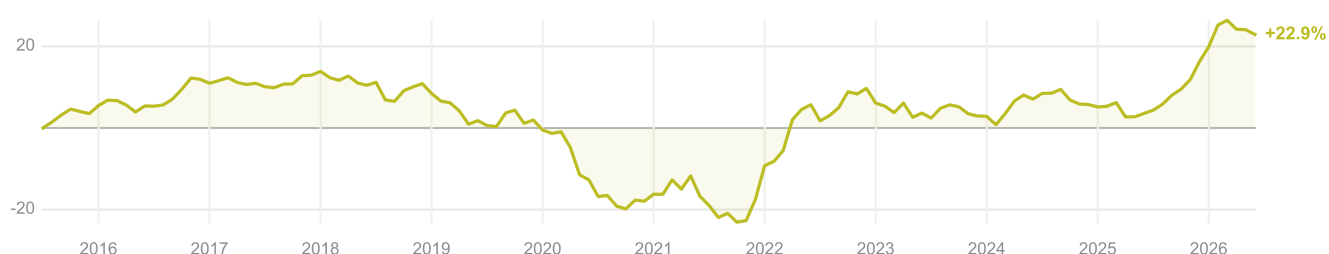
Earnings Quality

CAGR +6.0%

**119 Asset Growth - 1 Year**

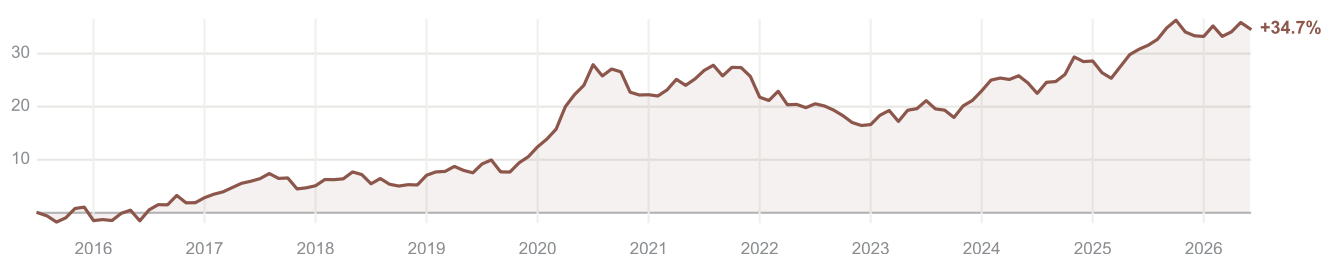
Capital Efficiency

CAGR +6.0%

**120 3Y Chg in Operating Cash Flow per Share**

Historical Growth

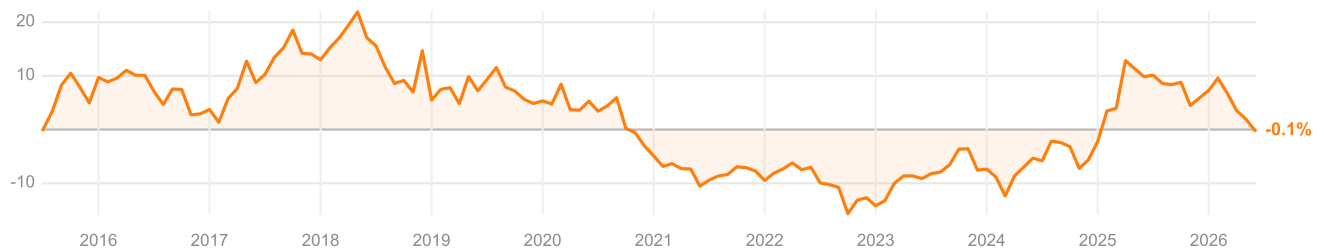
CAGR +6.0%



121 ● Adjusted 6 Month Average Share Turn Over

Price Momentum

CAGR +6.0%



122 ● Adj Growth Flow to Price

Valuation

CAGR +6.0%



123 ● 1Y Chg in Operating Cash Flow per Share

Historical Growth

CAGR +6.0%



124 ● Operating Margin

Earnings Quality

CAGR +5.9%



125 ● Adjusted Interest Coverage Ratio

Capital Efficiency

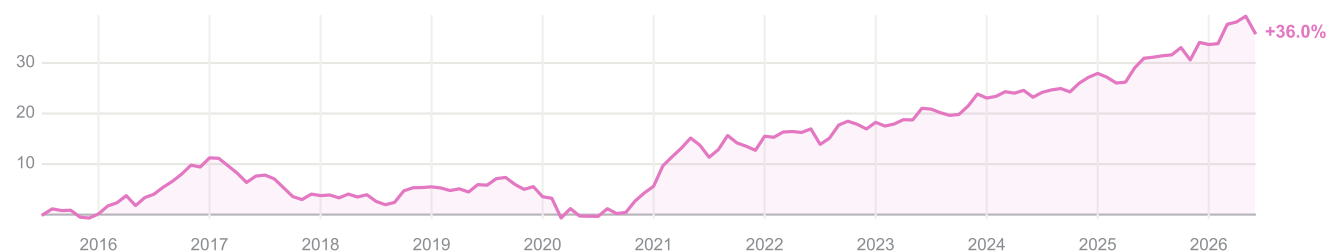
CAGR +5.8%



126 ● Adjusted Accruals

Earnings Quality

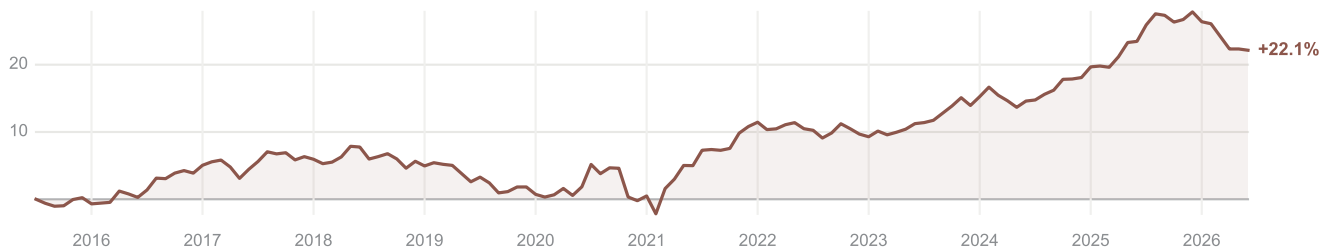
CAGR +5.8%



127 ● 3Y Chg in Free Cash Flow to Price

Historical Growth

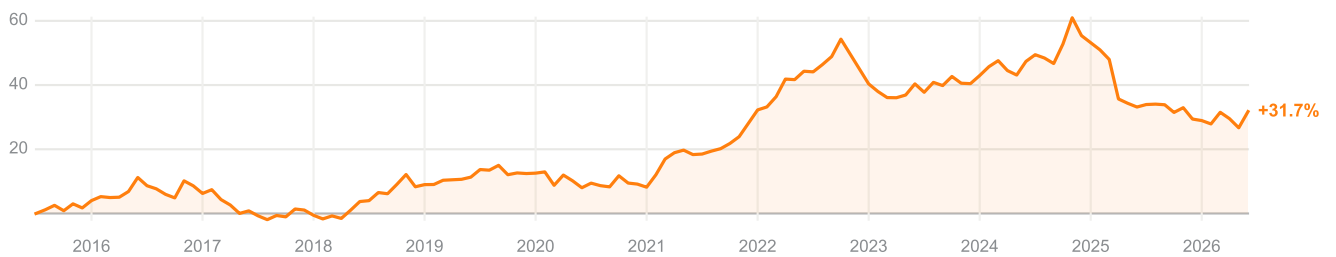
CAGR +5.7%



128 ● Ownership Breadth Stability of Institution Ownership by Buyers

Price Momentum

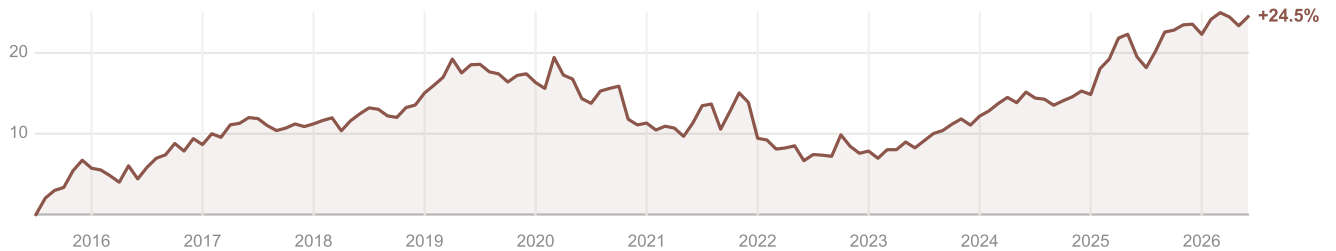
CAGR +5.7%



129 ● Dividend per Share Growth

Historical Growth

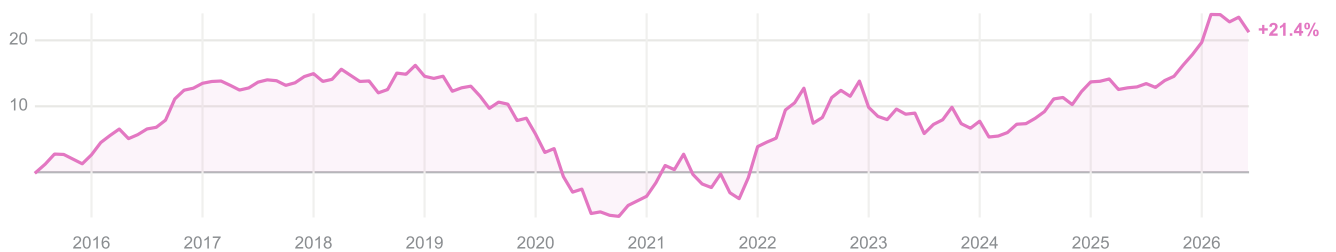
CAGR +5.7%



130 ● Accrual Ratio - Balance Sheet

Earnings Quality

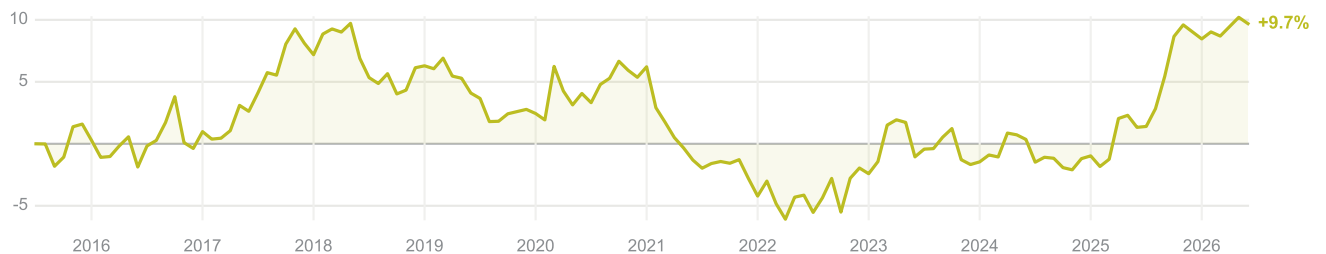
CAGR +5.6%



131 ● Long Term Debt to Assets Ratio

Capital Efficiency

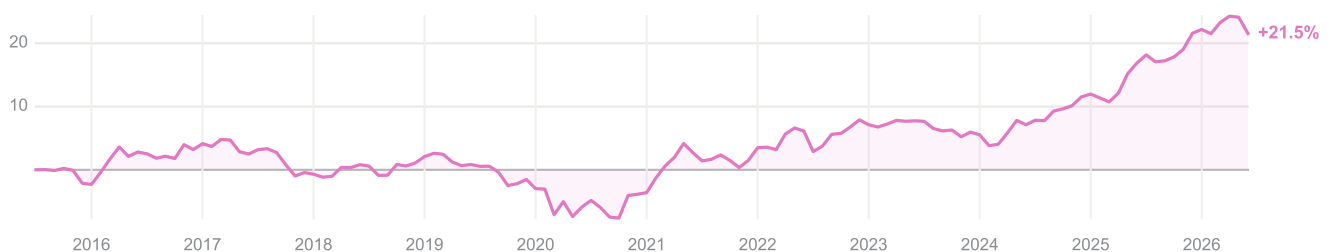
CAGR +5.6%

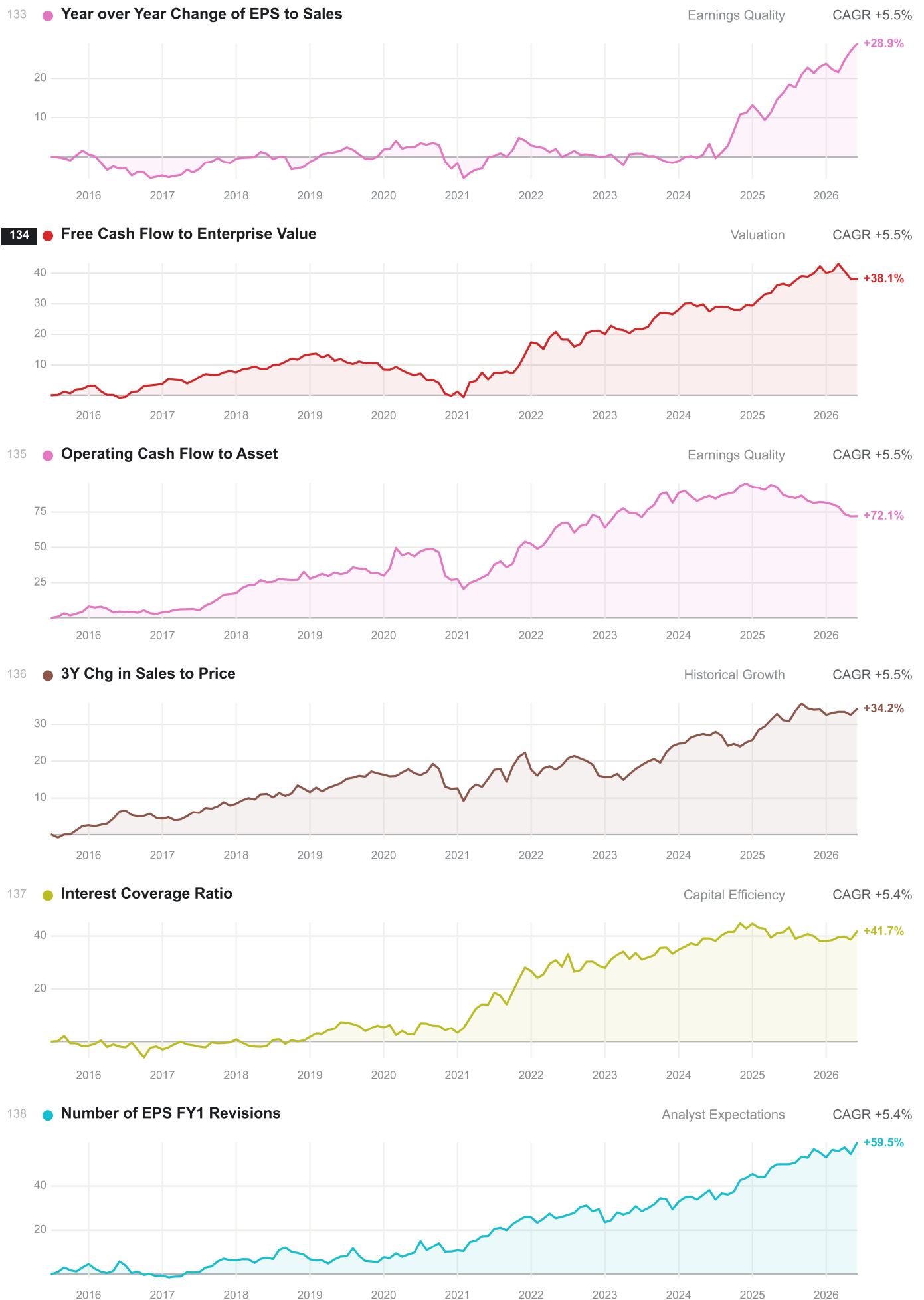


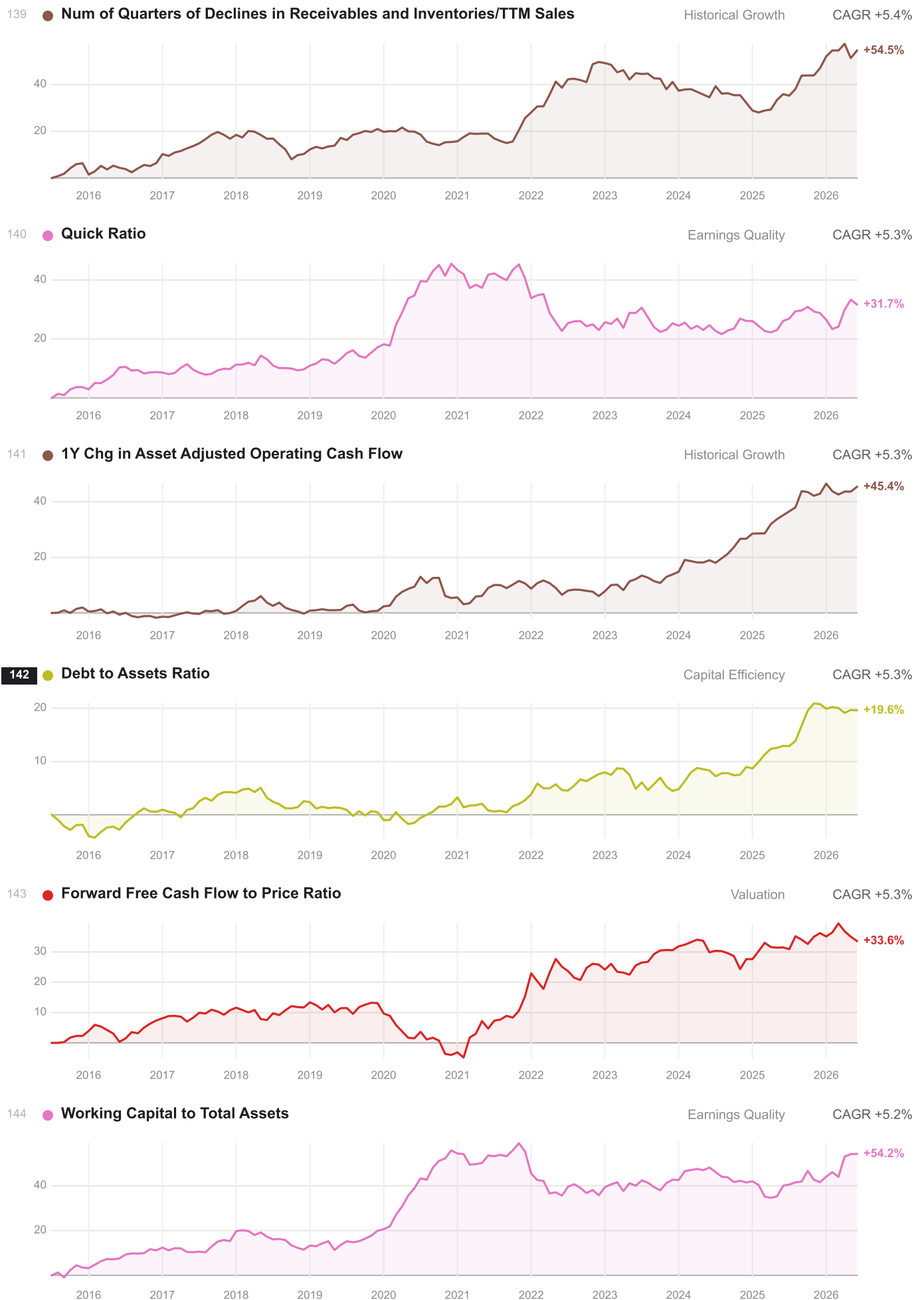
132 ● Accrual Ratio - Cash Flows

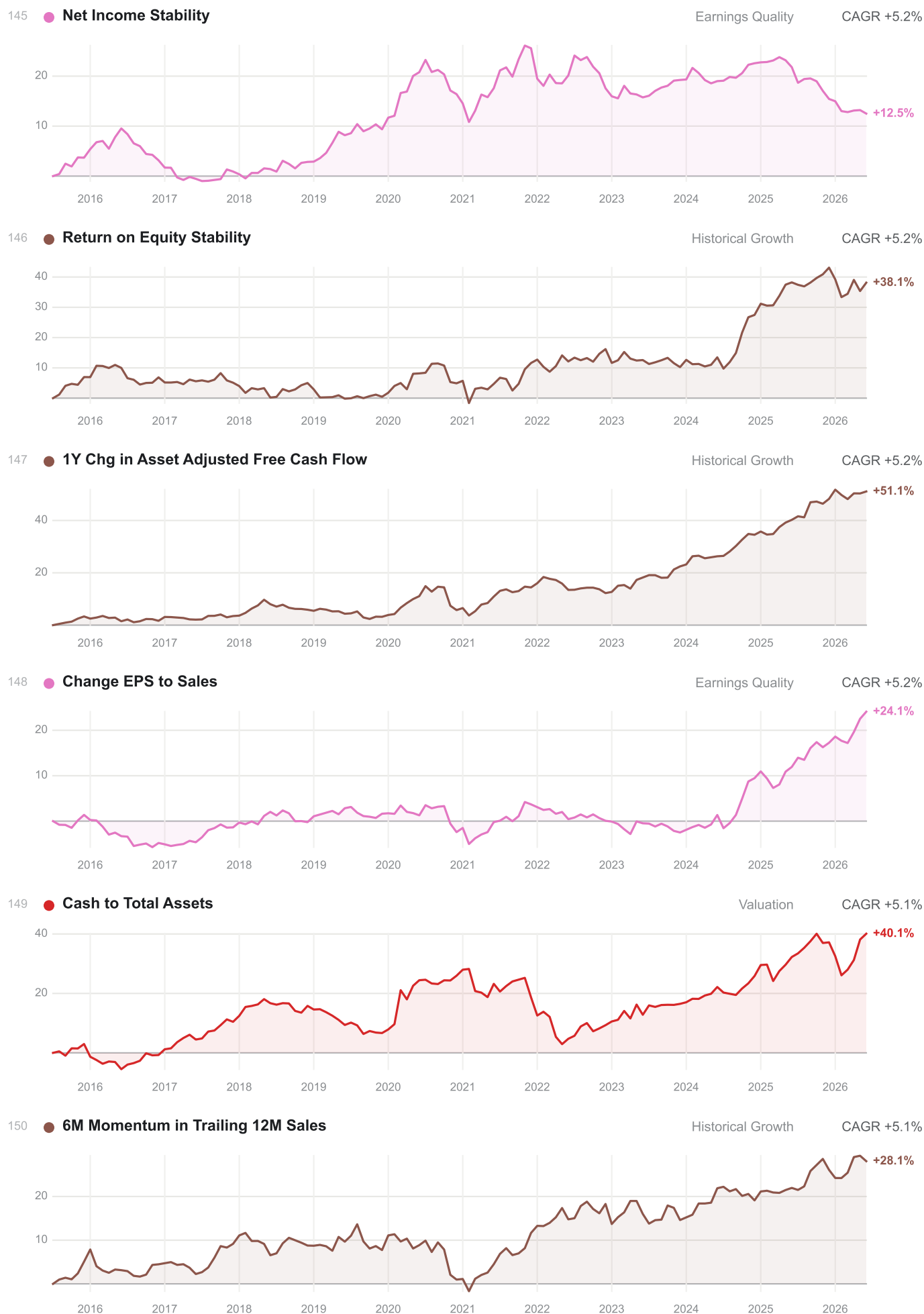
Earnings Quality

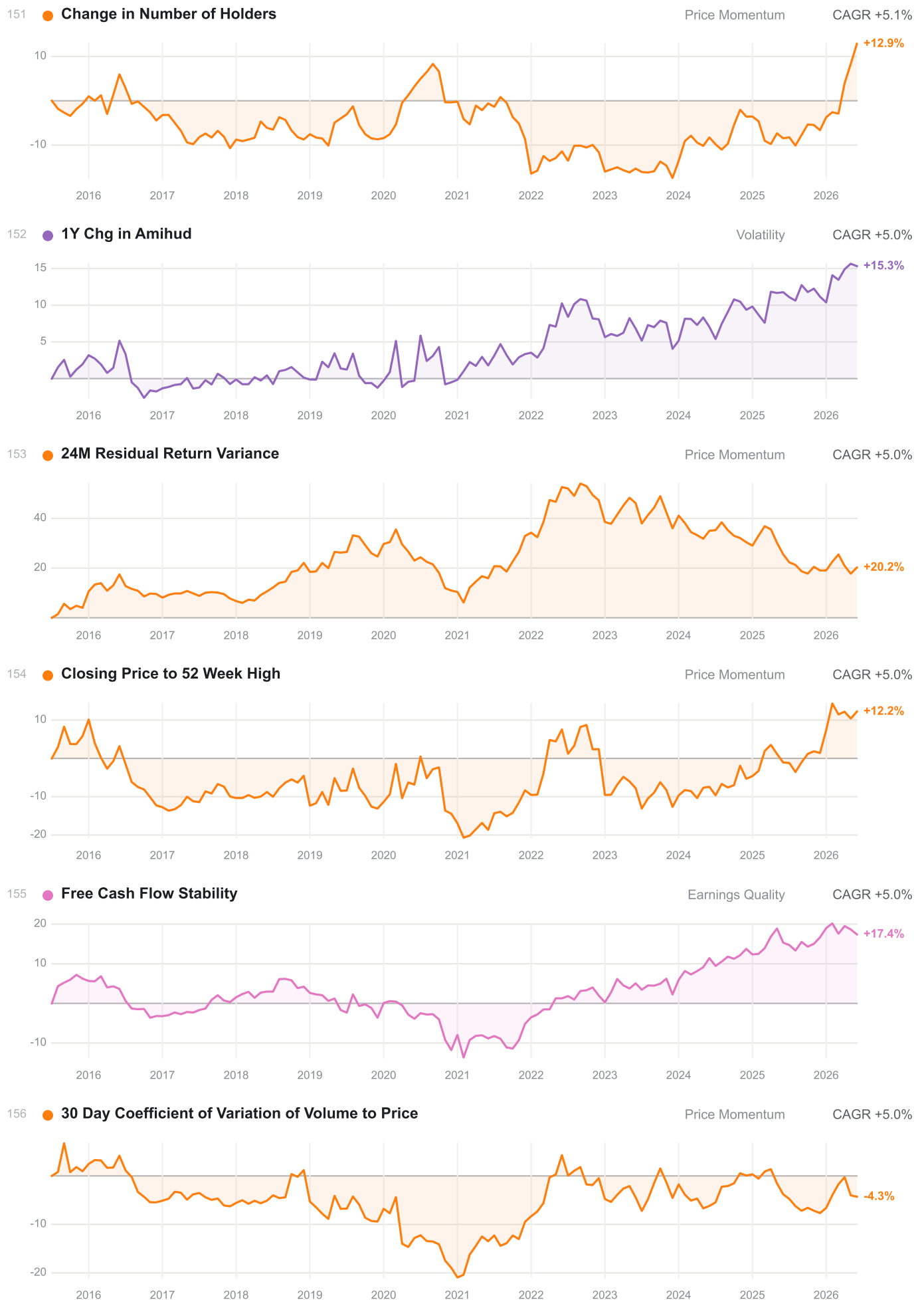
CAGR +5.6%

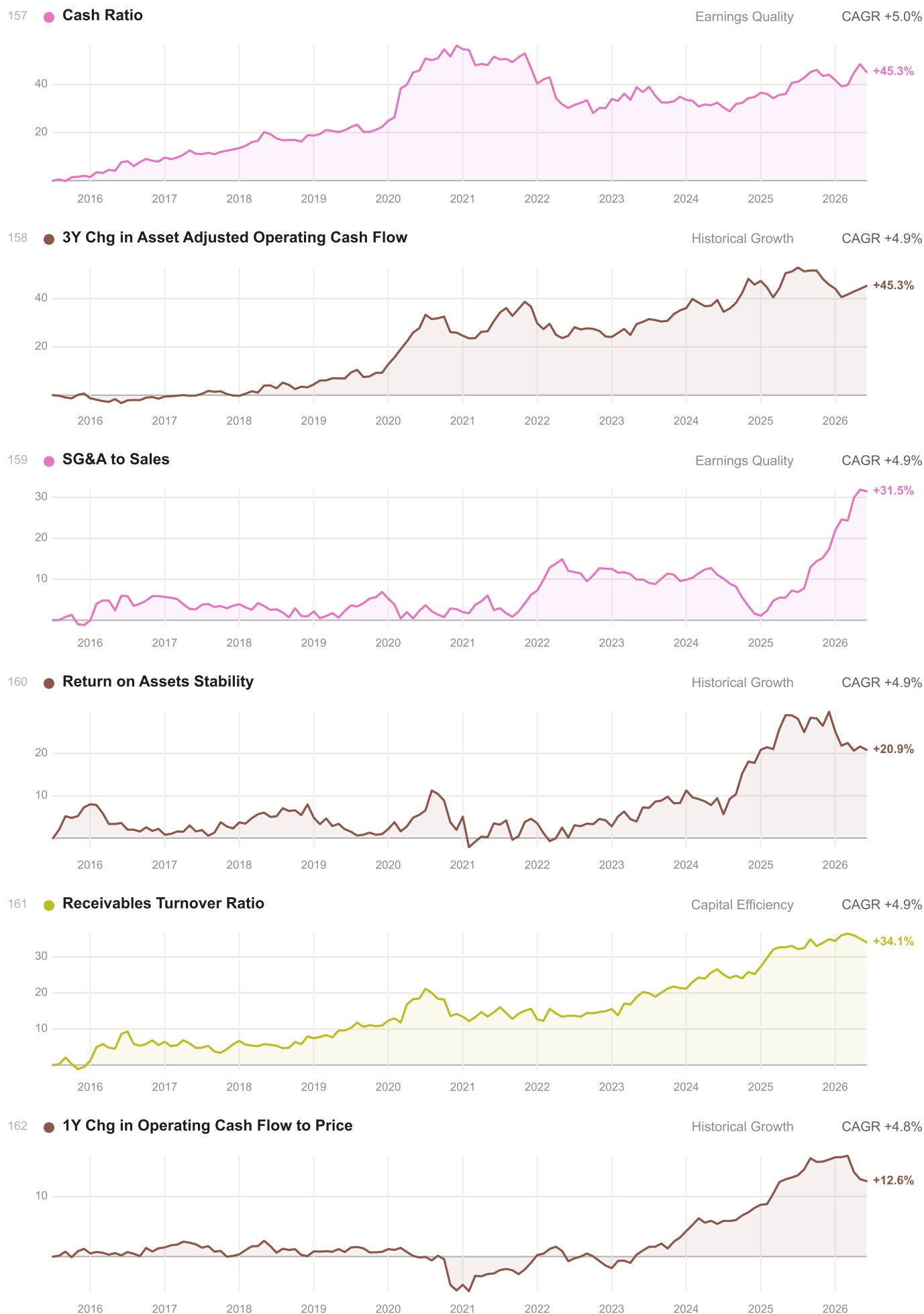








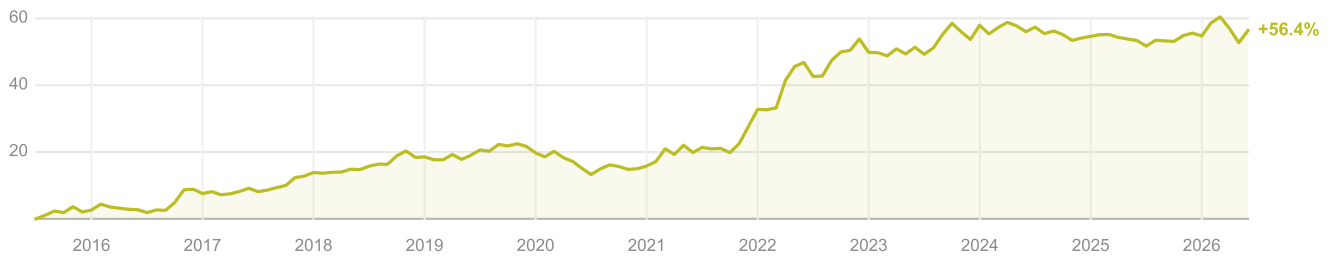




163 ● 1Y Chg in Shares Outstanding

Capital Efficiency

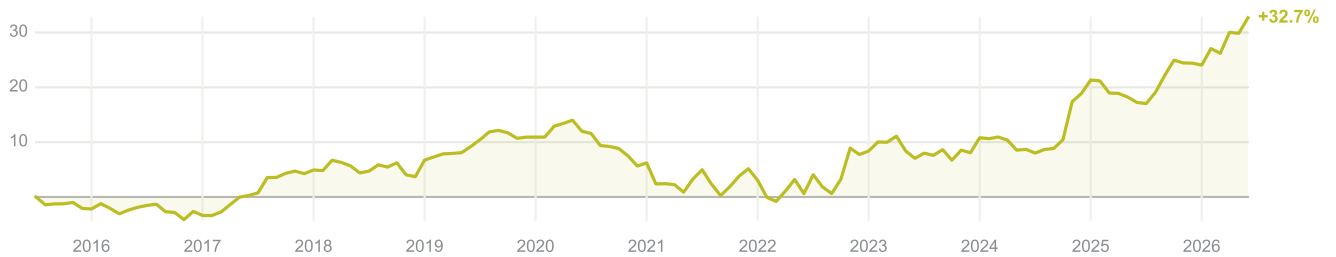
CAGR +4.8%



164 ● 1Y Chg in Stock Buybacks

Capital Efficiency

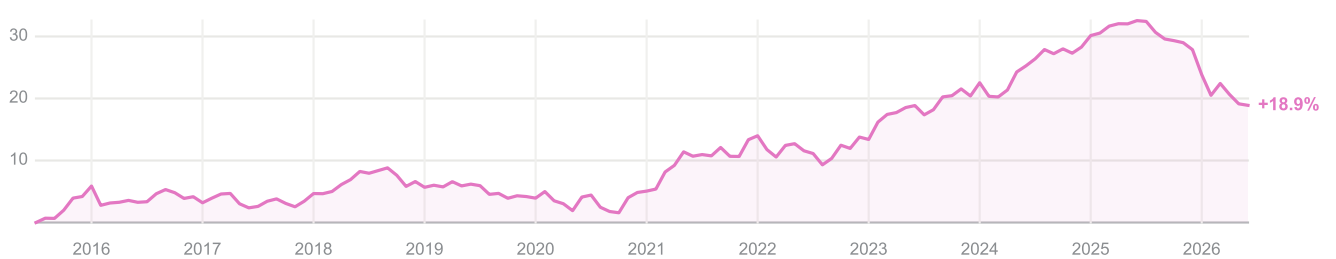
CAGR +4.8%



165 ● Depreciation to Capex Ratio

Earnings Quality

CAGR +4.8%



166 ● Free Cash Flow to Price

Valuation

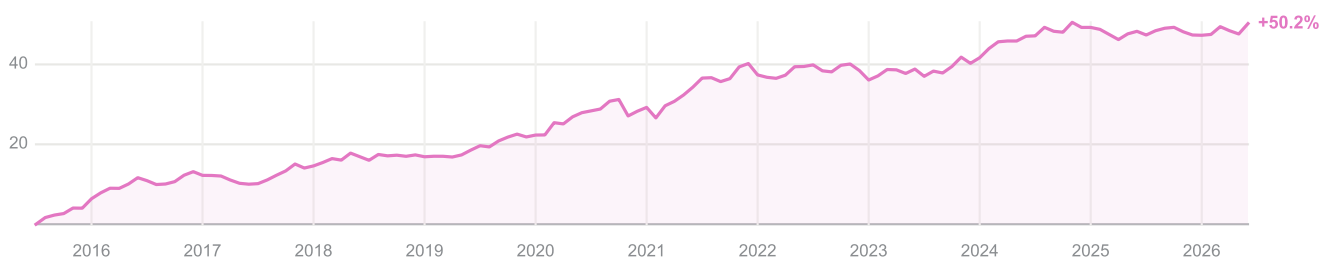
CAGR +4.7%



167 ● Asset Turnover

Earnings Quality

CAGR +4.7%



168 ● 1Y Chg in Free Cash Flow to Price

Historical Growth

CAGR +4.6%



169 ● **Adj Forward Free Cash Flow to Price Ratio**

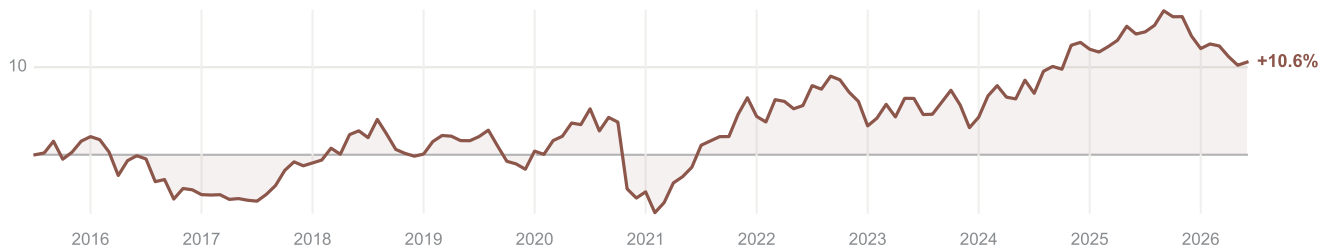
Valuation

CAGR +4.6%

170 ● **EPS Stability**

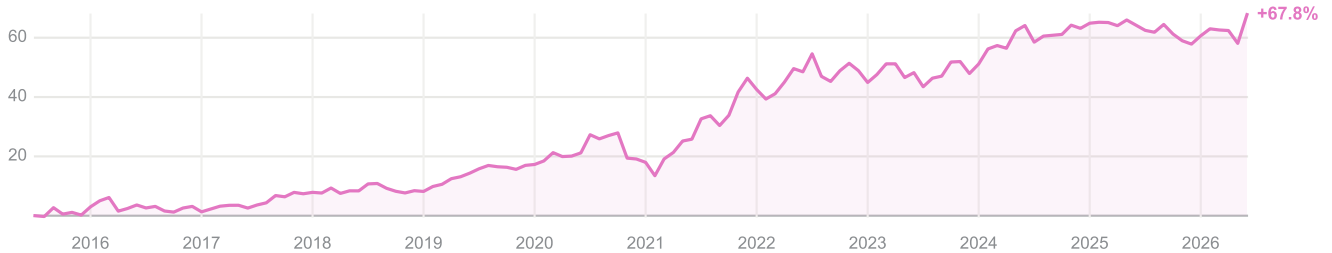
Historical Growth

CAGR +4.6%

171 ● **Cash Flow to Invested Capital**

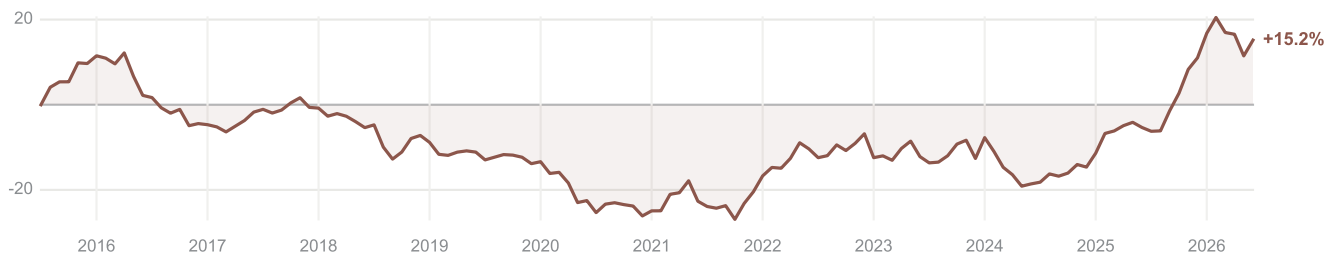
Earnings Quality

CAGR +4.6%

172 ● **1Y Chg in Sales Turnover**

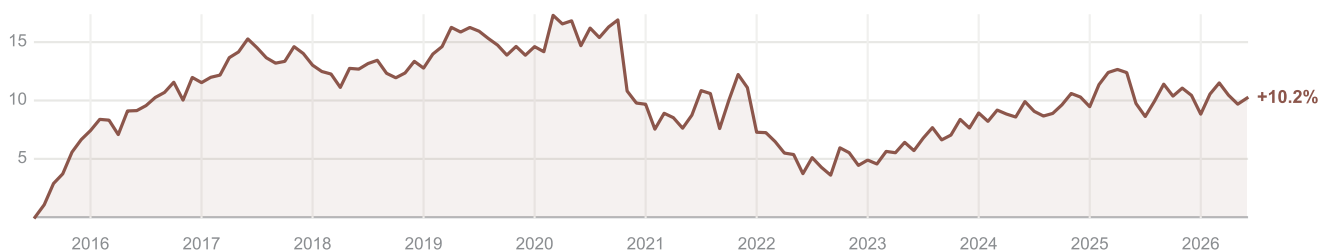
Historical Growth

CAGR +4.6%

173 ● **Dividend Growth**

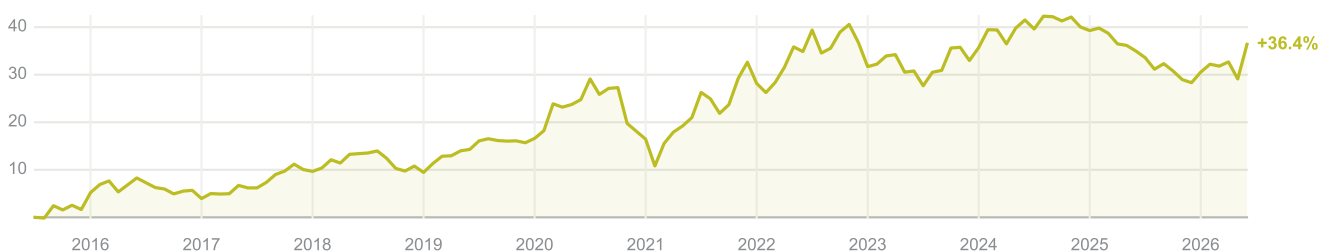
Historical Growth

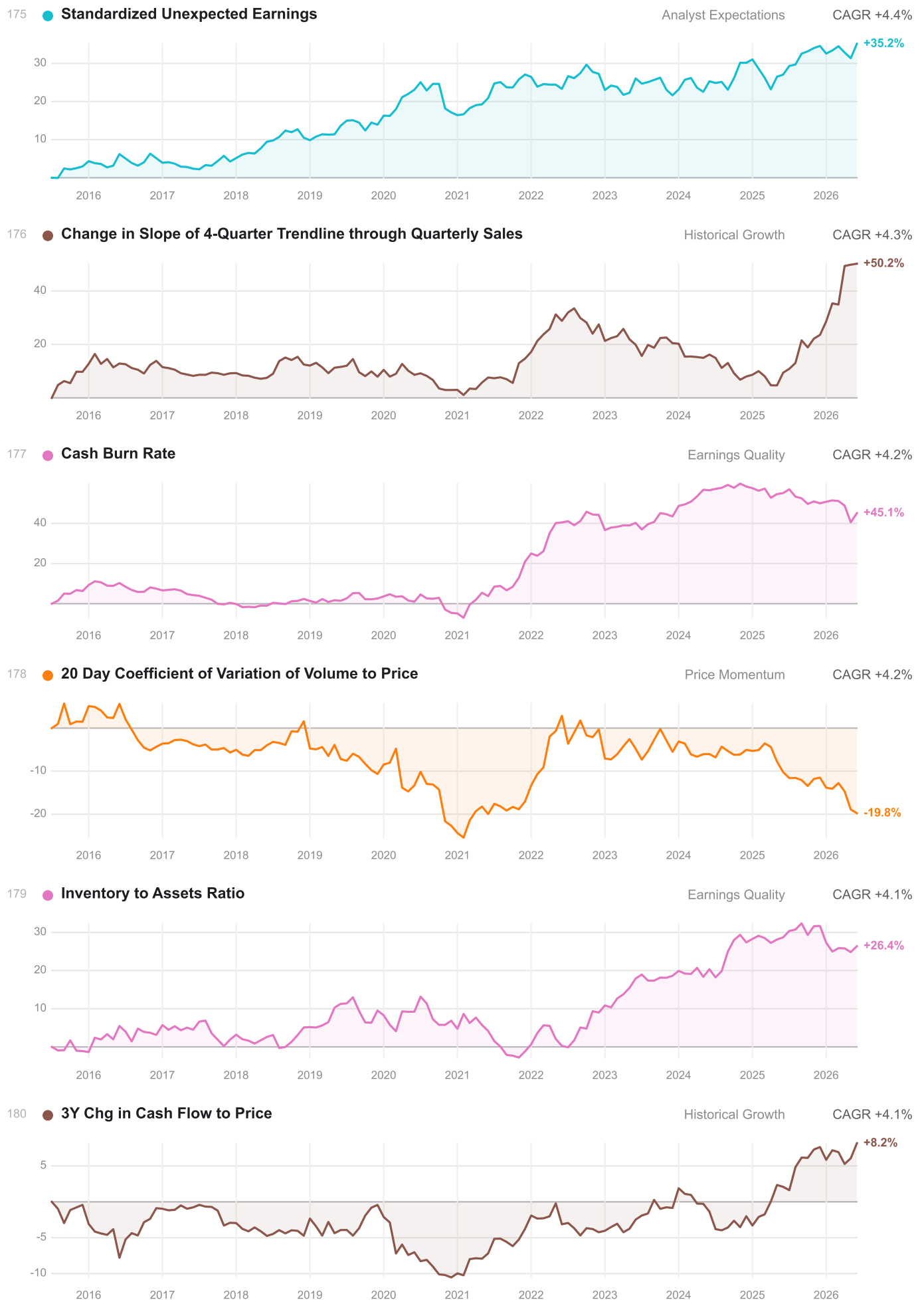
CAGR +4.5%

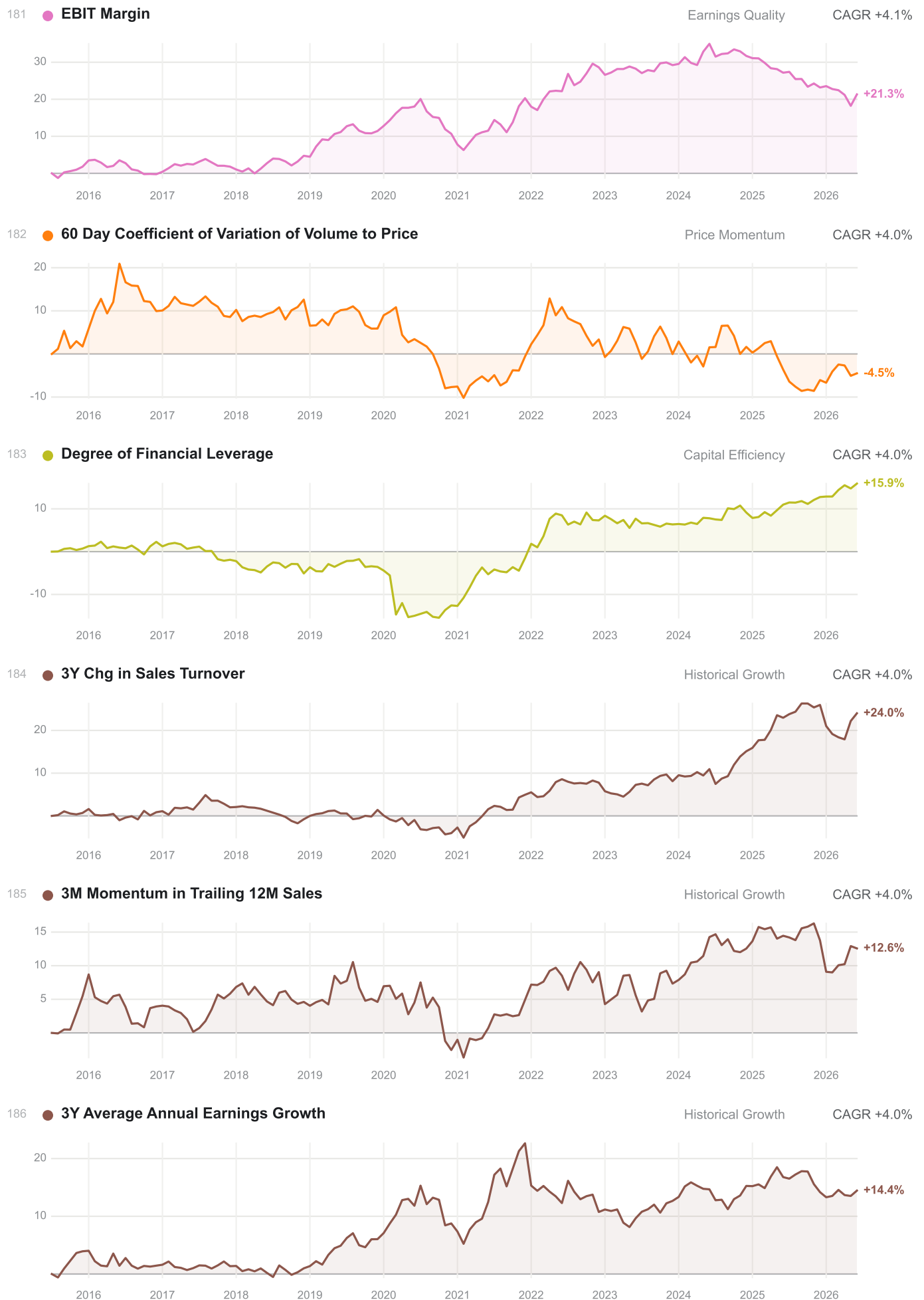
174 ● **Return on Invested Capital**

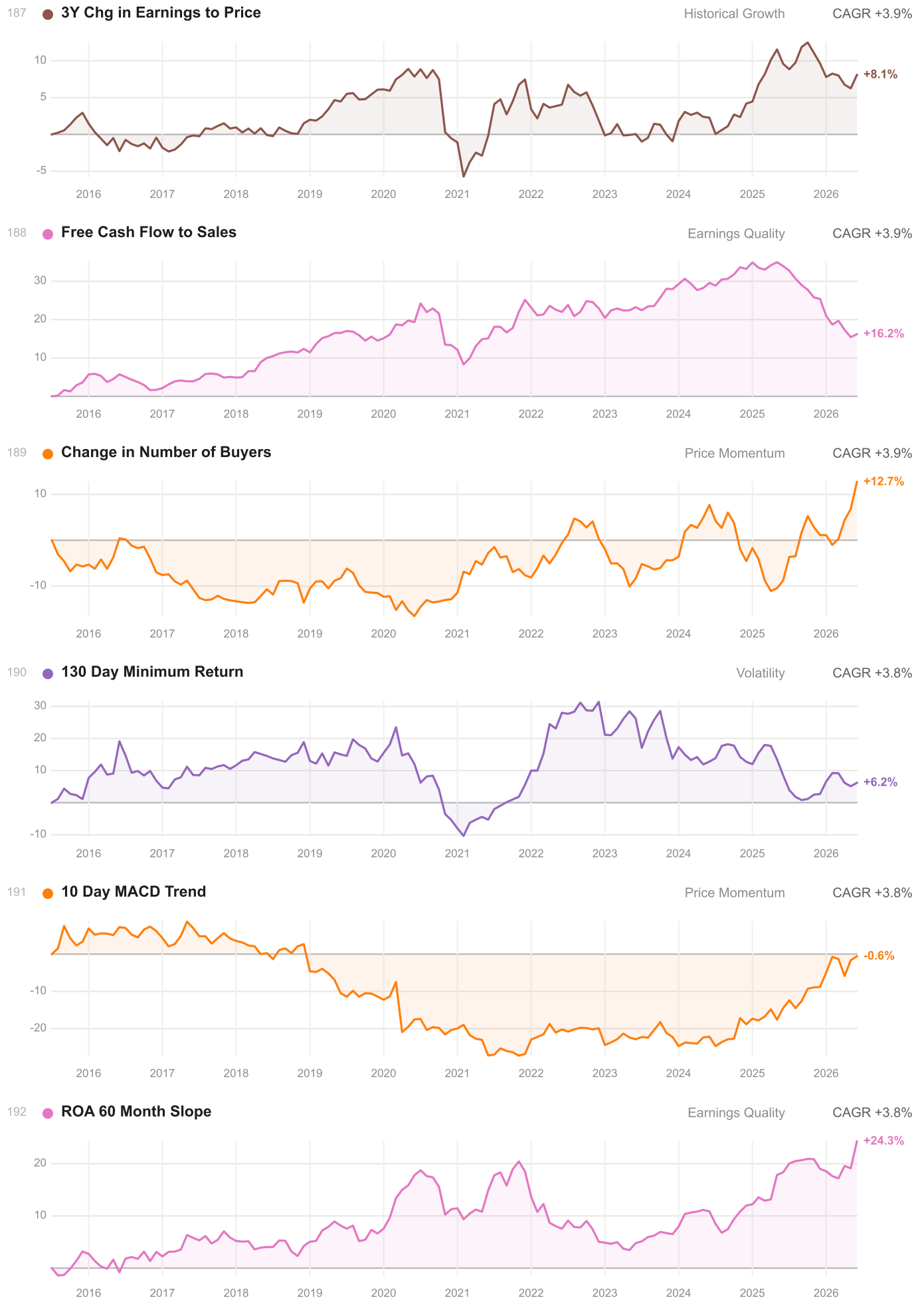
Capital Efficiency

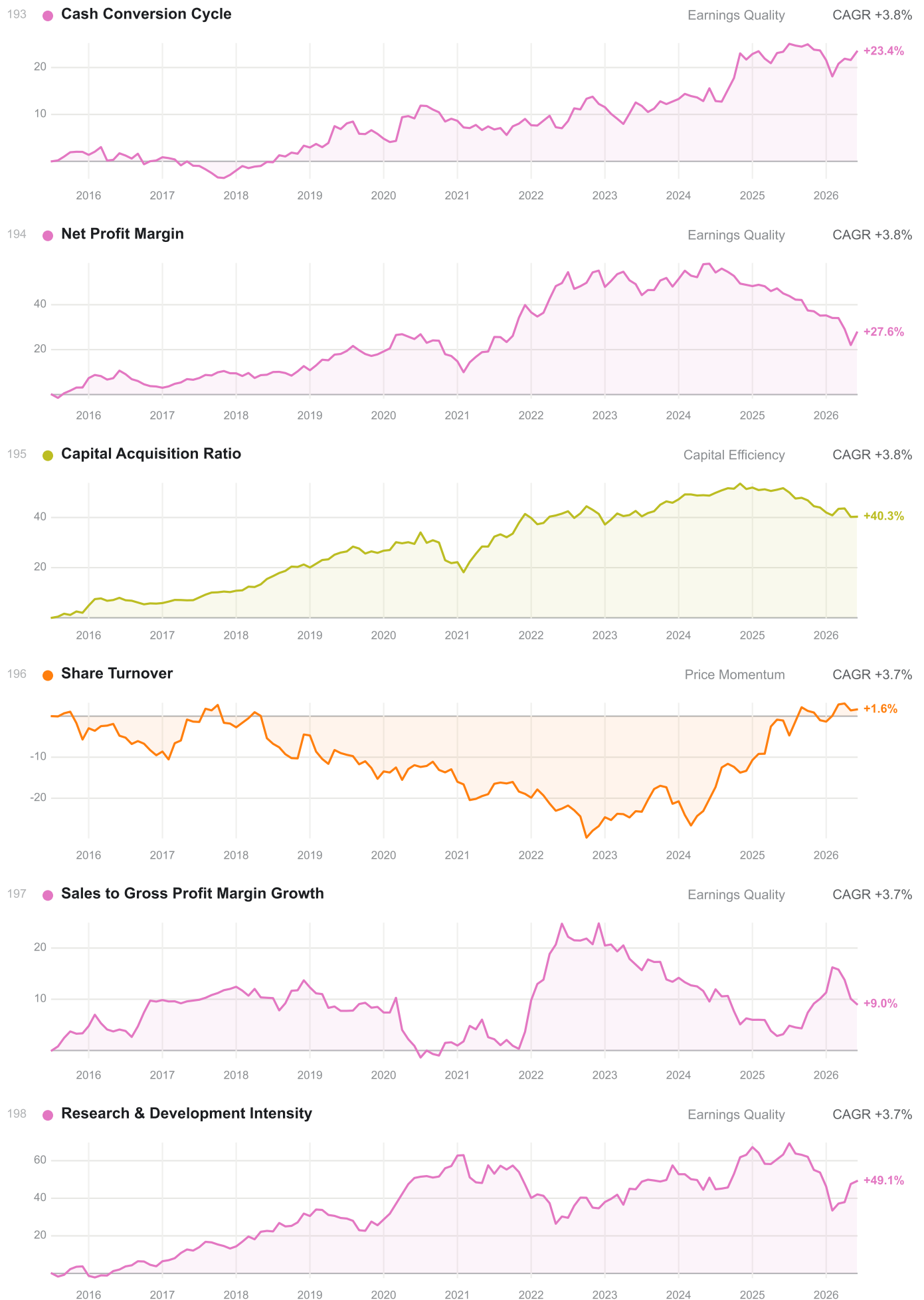
CAGR +4.5%

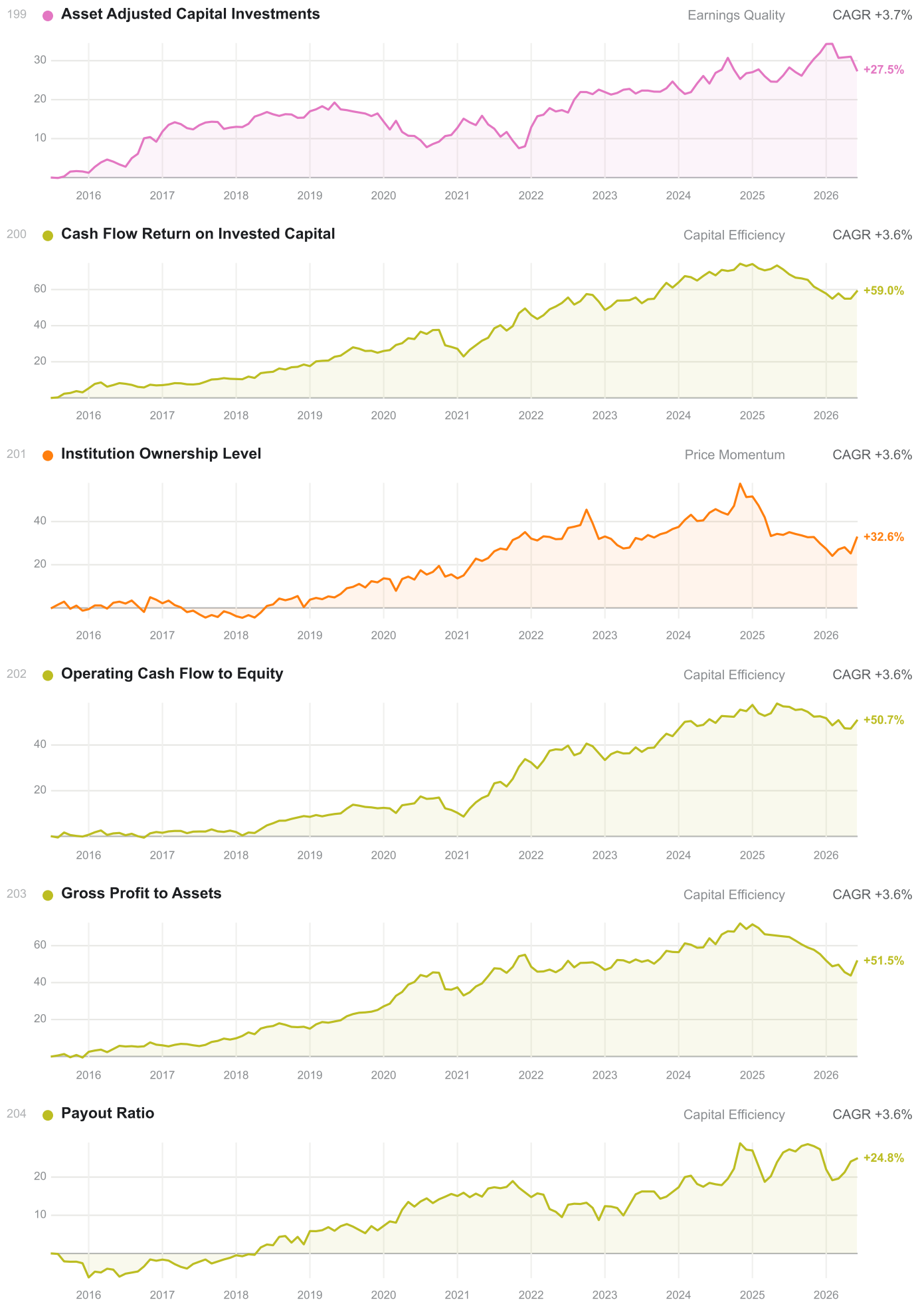


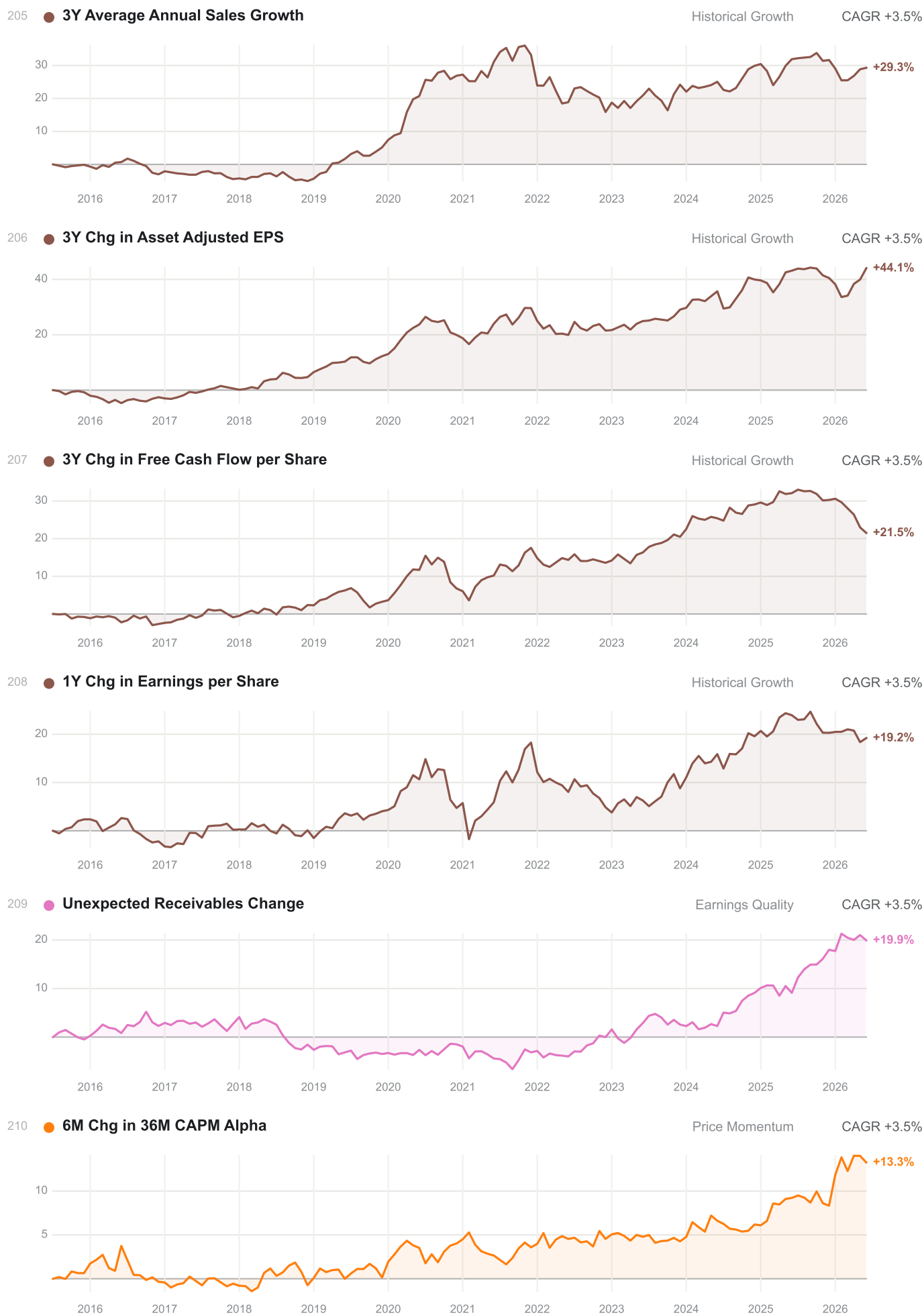








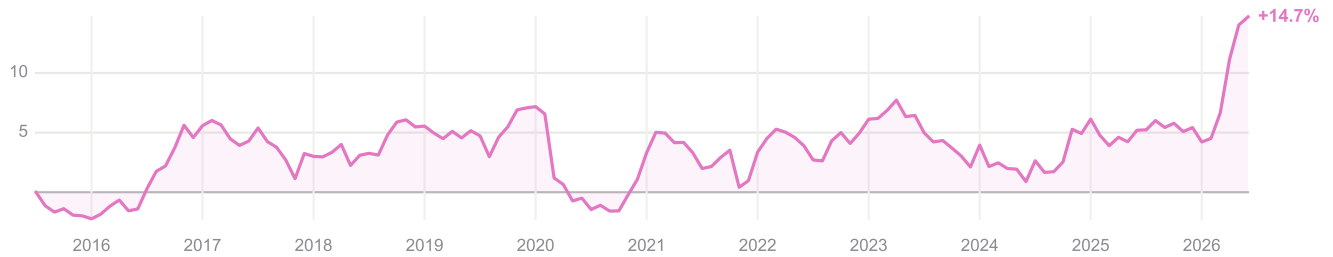




211 ● 1Y Chg in Sales to Earnings

Earnings Quality

CAGR +3.5%



212 ● Operating Cash Flow Stability

Earnings Quality

CAGR +3.4%



213 ● 6 Month Average Share Turn Over

Price Momentum

CAGR +3.4%



214 ● EBIT to Price

Valuation

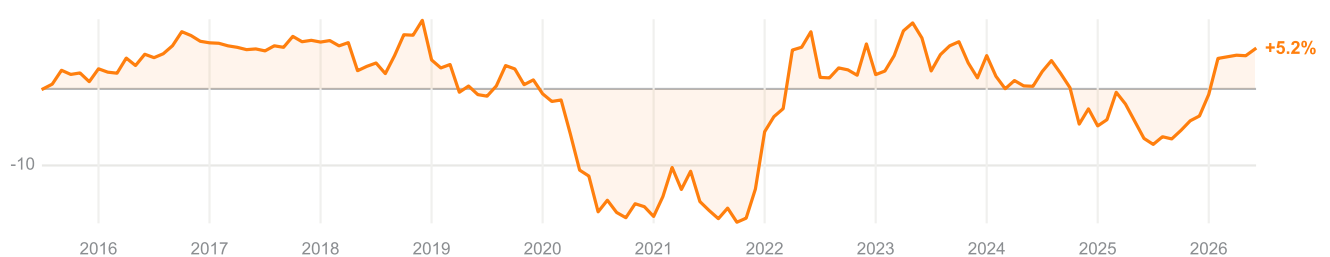
CAGR +3.4%



215 ● Ownership Breadth Stability of Institution Ownership by Holders

Price Momentum

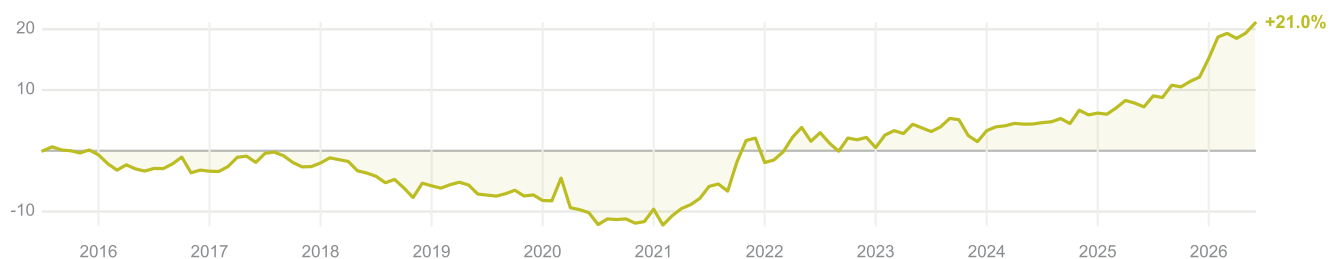
CAGR +3.3%



216 ● Year over Year Change of Total Debt

Capital Efficiency

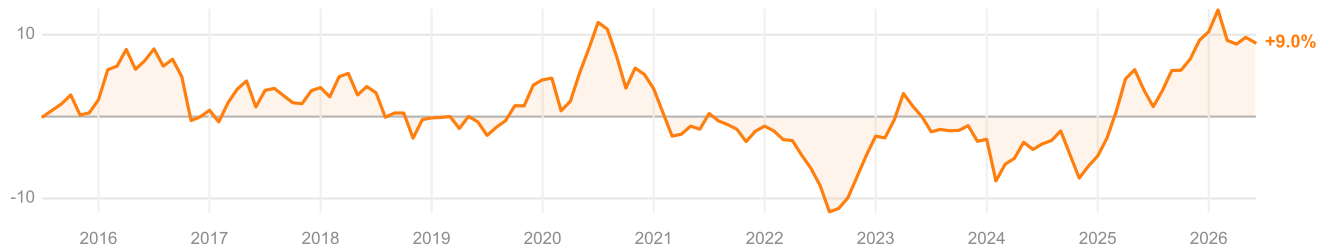
CAGR +3.3%



217 ● Foreign Institution Ownership

Price Momentum

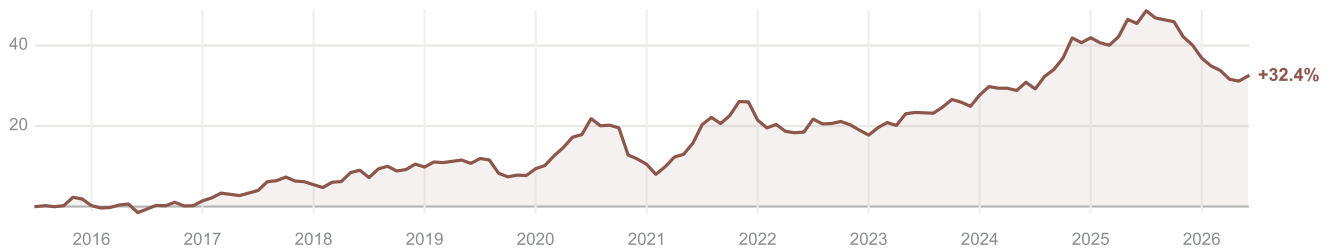
CAGR +3.3%



218 ● 3Y Chg in Asset Adjusted Free Cash Flow

Historical Growth

CAGR +3.2%



219 ● Free Cash Flow to Common Equity

Valuation

CAGR +3.2%



220 ● Sustainable Growth Rate

Historical Growth

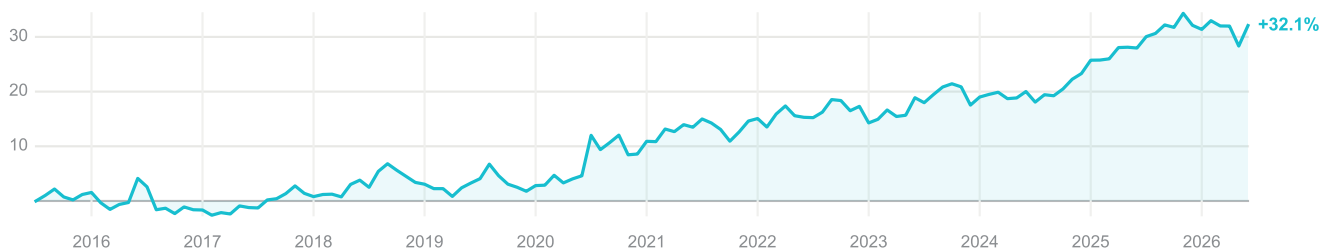
CAGR +3.1%



221 ● Adjusted Number of EPS FY1 Revisions

Analyst Expectations

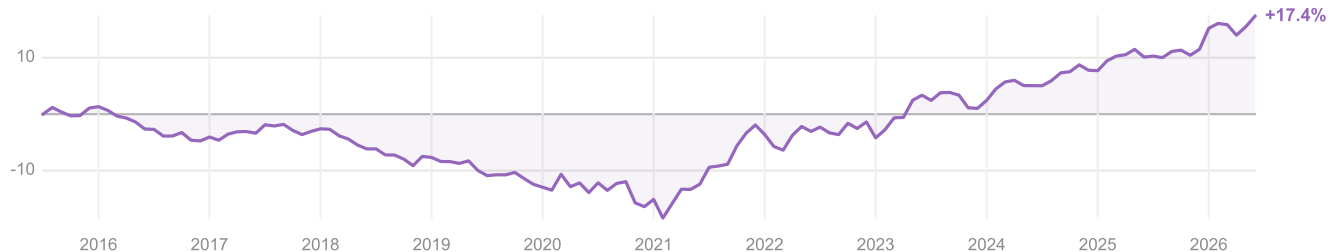
CAGR +3.1%



222 ● 1Y Chg in Debt to Assets

Volatility

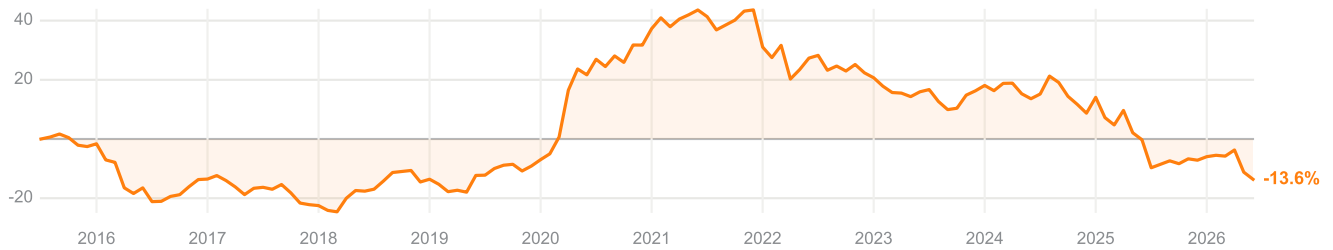
CAGR +3.0%



223 ● **Standardized 36M Residual**

Price Momentum

CAGR +2.9%

224 ● **1Y Chg in Cash Flow per Share**

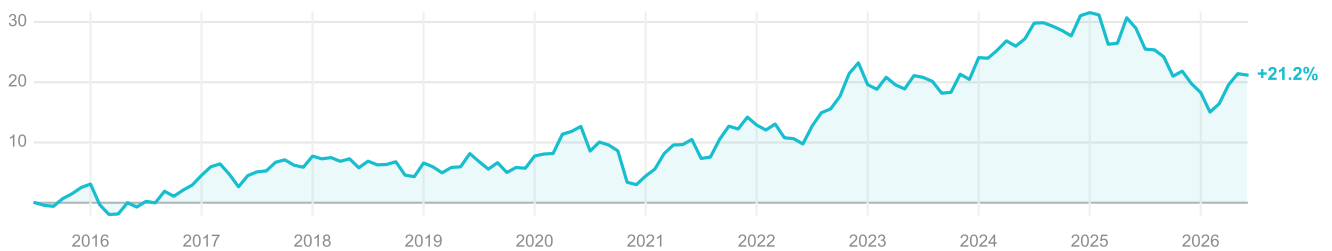
Historical Growth

CAGR +2.8%

225 ● **Target Price Gap to 6M EMA**

Analyst Expectations

CAGR +2.8%

226 ● **3Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +2.8%

227 ● **Net Current Assets to Price Ratio**

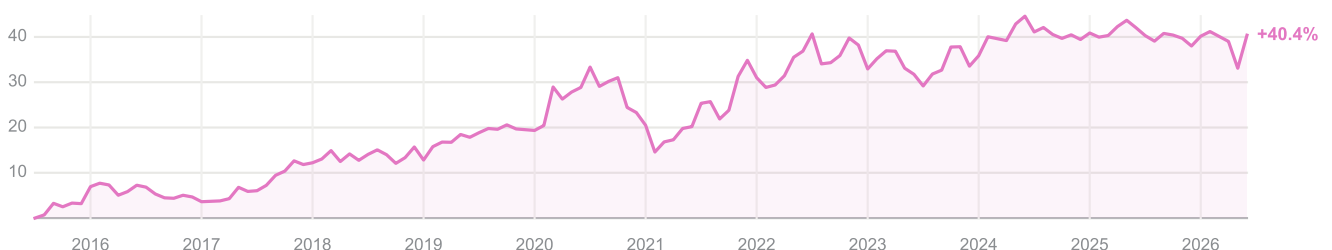
Valuation

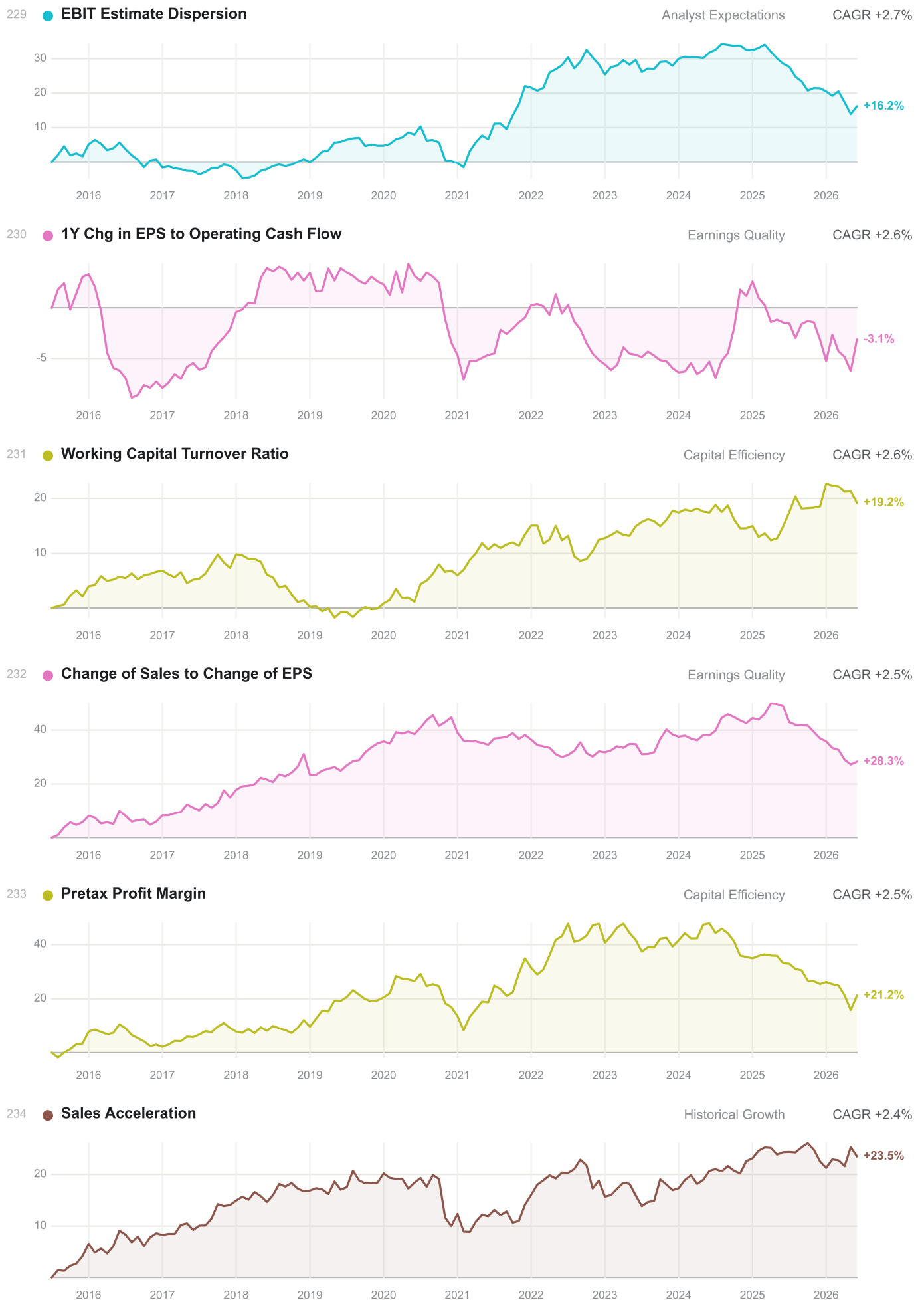
CAGR +2.7%

228 ● **Pretax Income to Net Operating Assets**

Earnings Quality

CAGR +2.7%

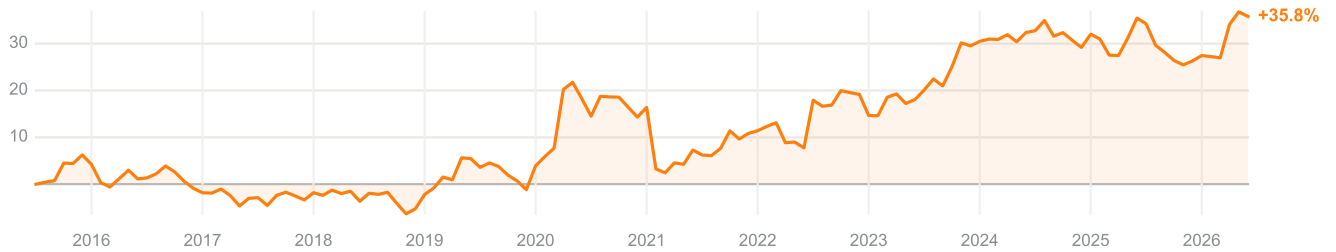




235 ● Risk Adjusted Relative Strength

Price Momentum

CAGR +2.4%



236 ● Analyst Earnings Estimate Diffusion

Analyst Expectations

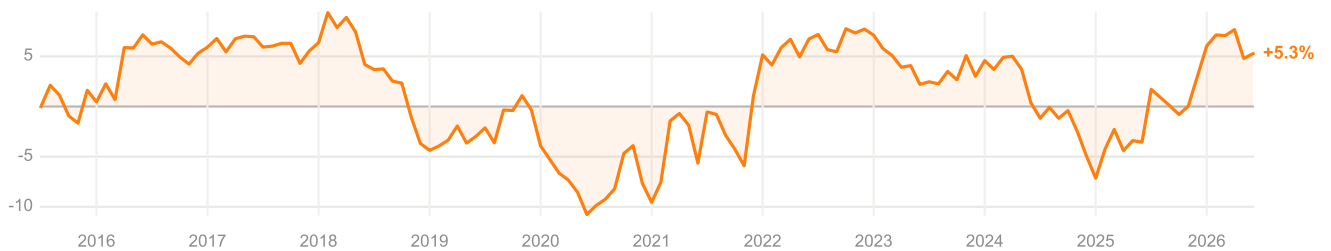
CAGR +2.4%



237 ● Change of Ownership Level - Hedge Fund

Price Momentum

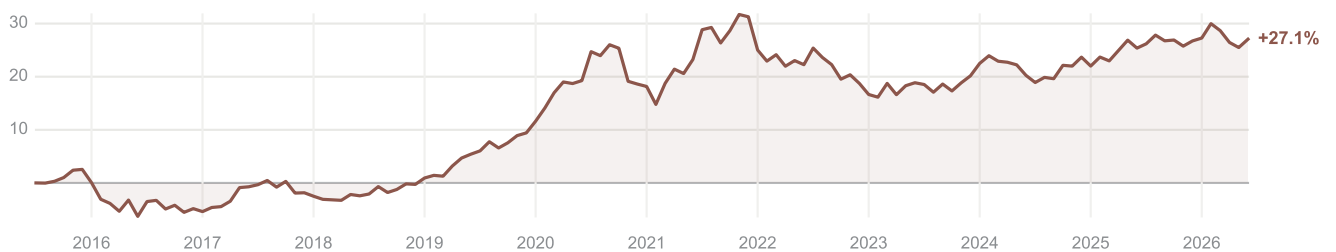
CAGR +2.4%



238 ● 3Y Chg in Cash Flow per Share

Historical Growth

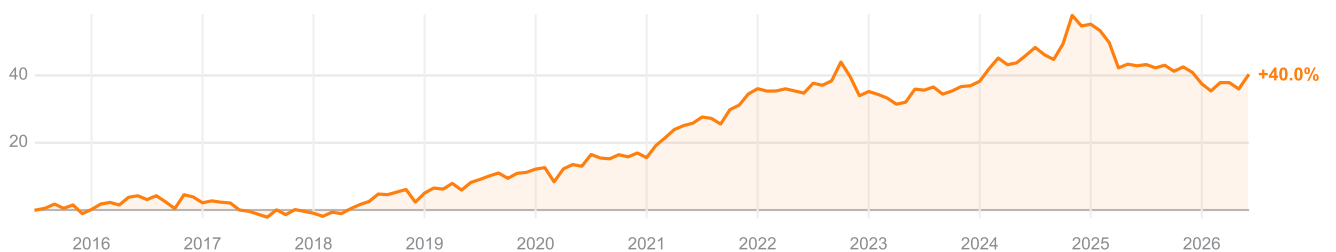
CAGR +2.3%



239 ● Investment Duration

Price Momentum

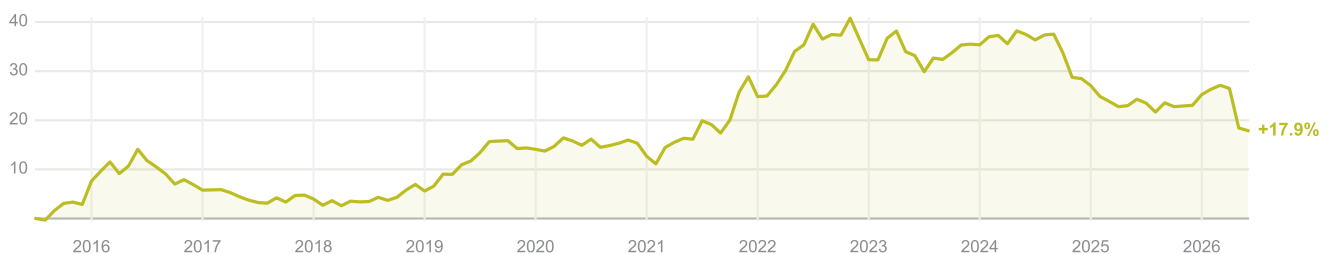
CAGR +2.2%



240 ● Normalized ROE

Capital Efficiency

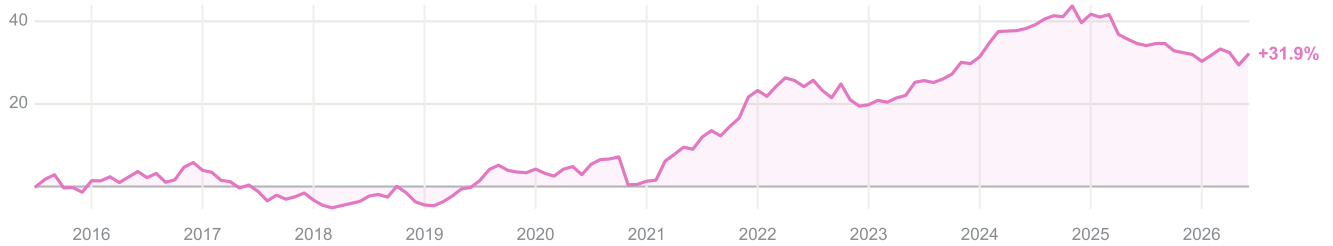
CAGR +2.1%



241 ● **Cash to Sales**

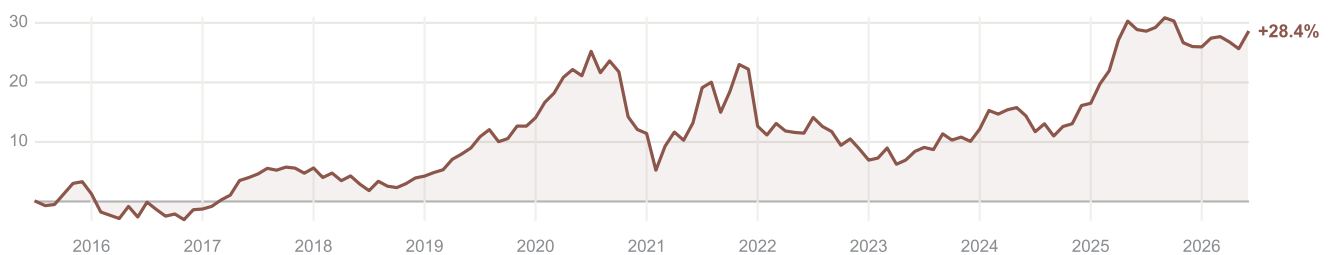
Earnings Quality

CAGR +2.1%

242 ● **3Y Chg in Earnings per Share**

Historical Growth

CAGR +2.0%

243 ● **Institution Ownership Turnover**

Price Momentum

CAGR +2.0%

244 ● **Max Daily Return in the Past 6 Months**

Price Momentum

CAGR +2.0%

245 ● **Revenue Estimate Dispersion**

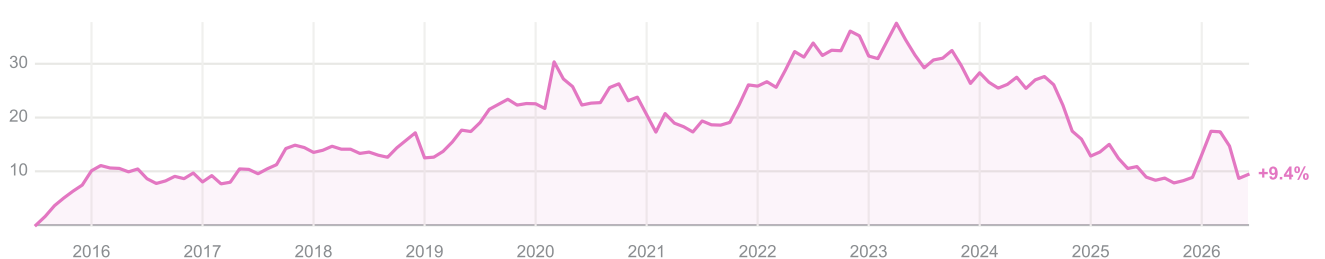
Analyst Expectations

CAGR +1.9%

246 ● **Retained Earnings to Total Assets**

Earnings Quality

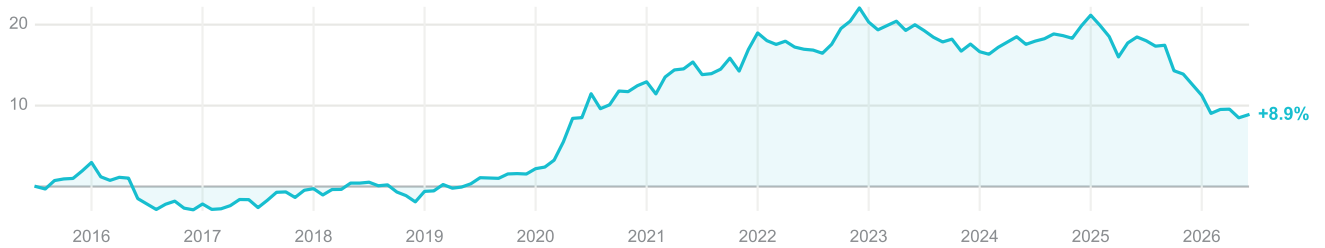
CAGR +1.9%



247 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

Analyst Expectations

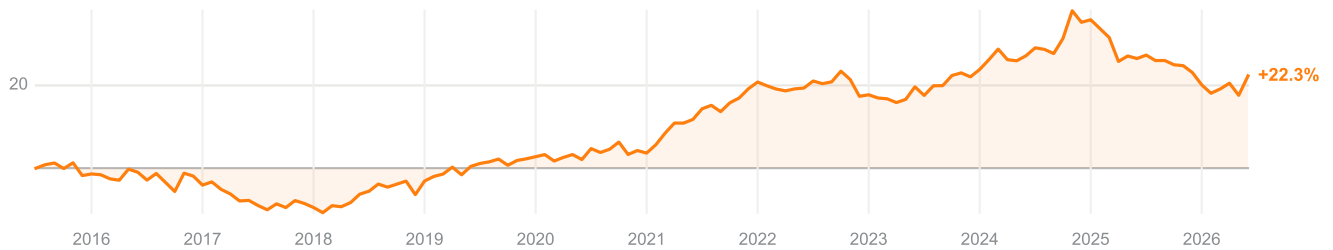
CAGR +1.9%



248 ● Institution Ownership Concentration

Price Momentum

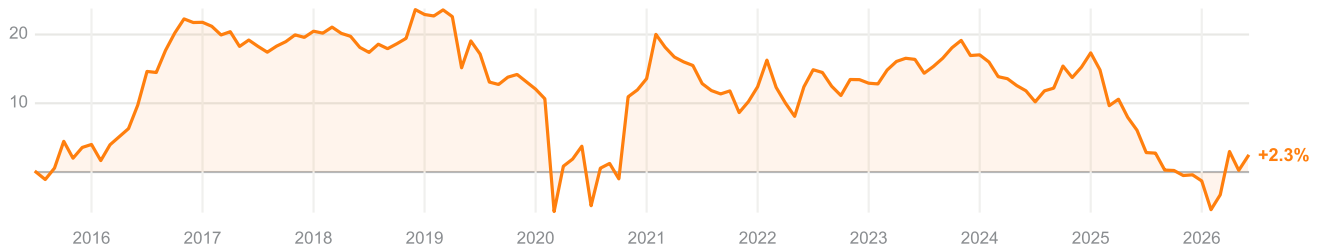
CAGR +1.8%



249 ● 5 Day Price Reversal

Price Momentum

CAGR +1.7%



250 ● 6M Chg in 18M CAPM Alpha

Price Momentum

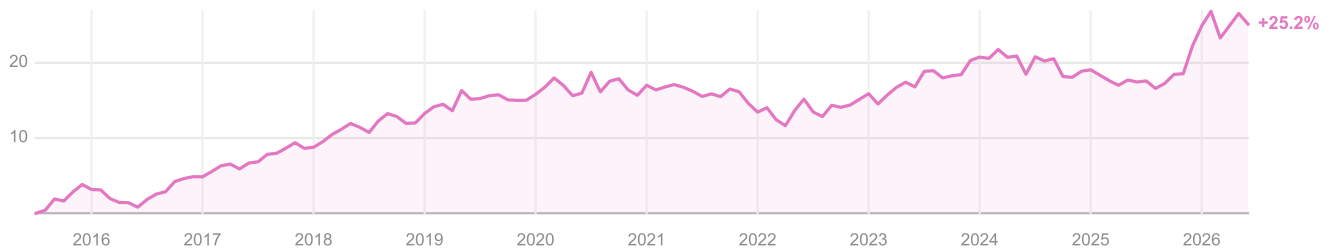
CAGR +1.6%



251 ● Year over Year Change of Inventory to Assets

Earnings Quality

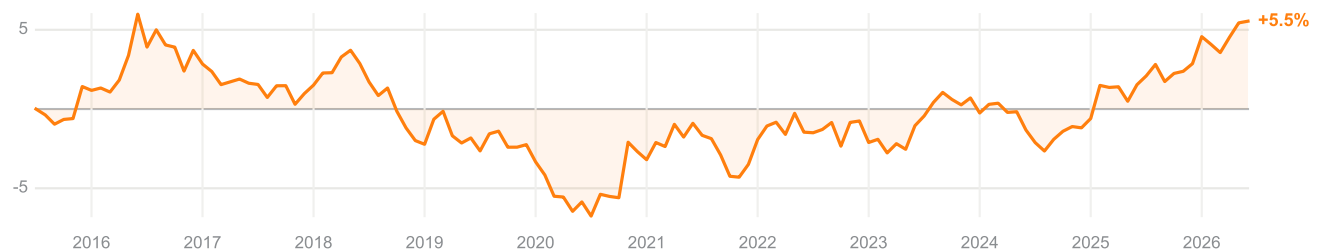
CAGR +1.6%

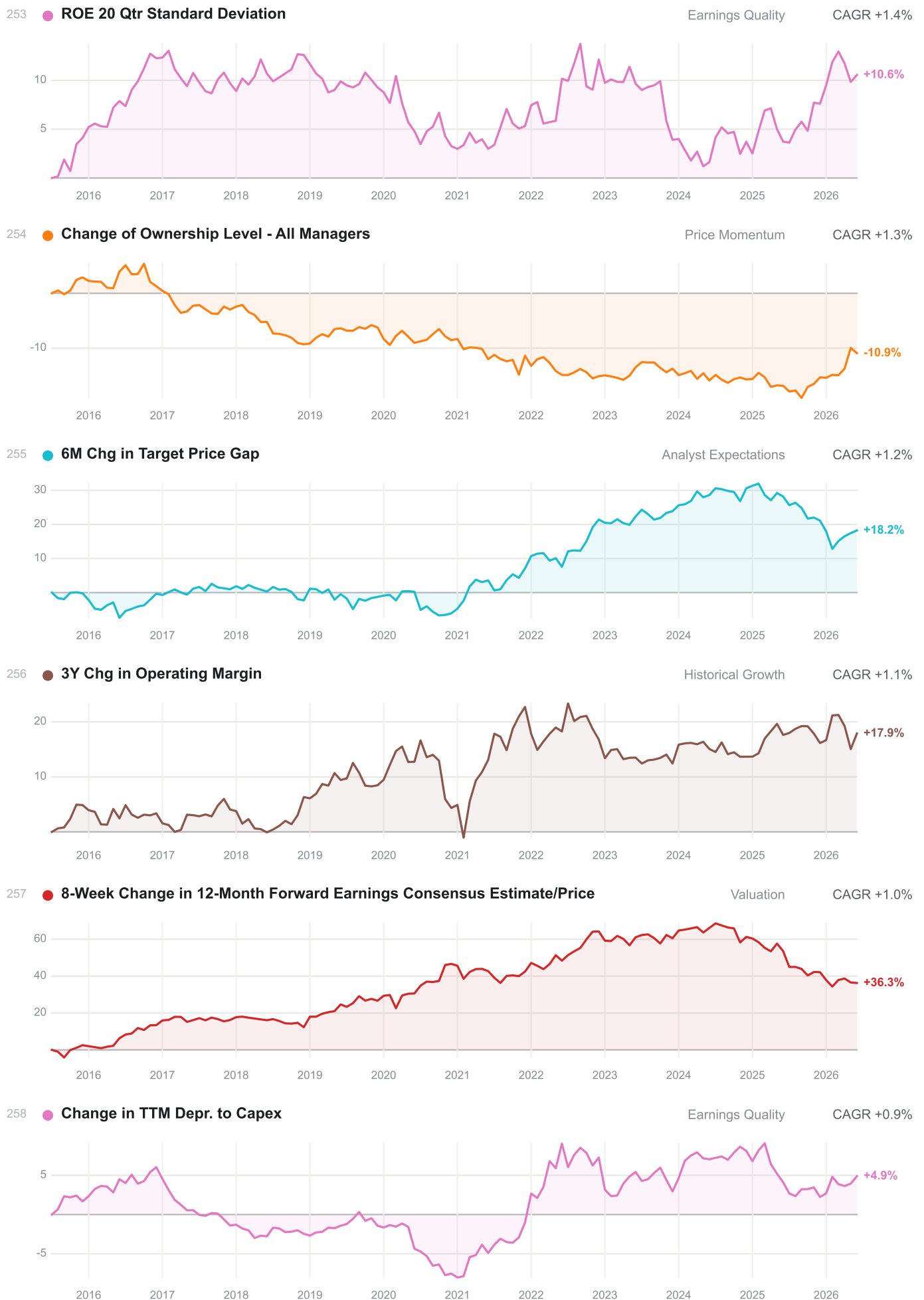


252 ● 6M Chg in 12M CAPM Alpha

Price Momentum

CAGR +1.4%

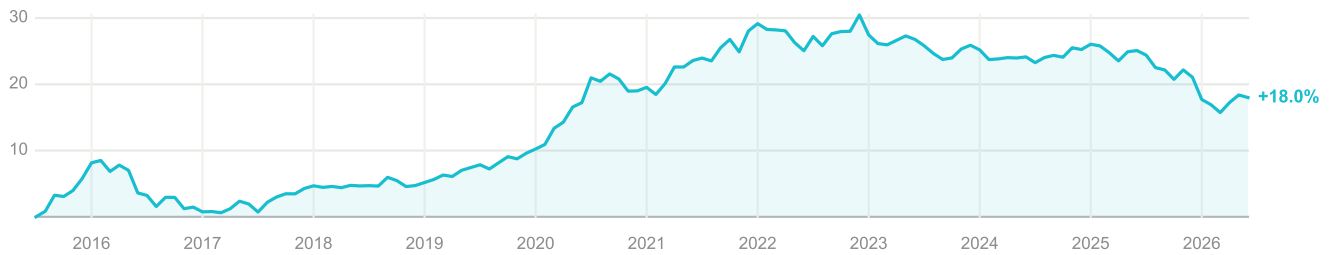




259 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

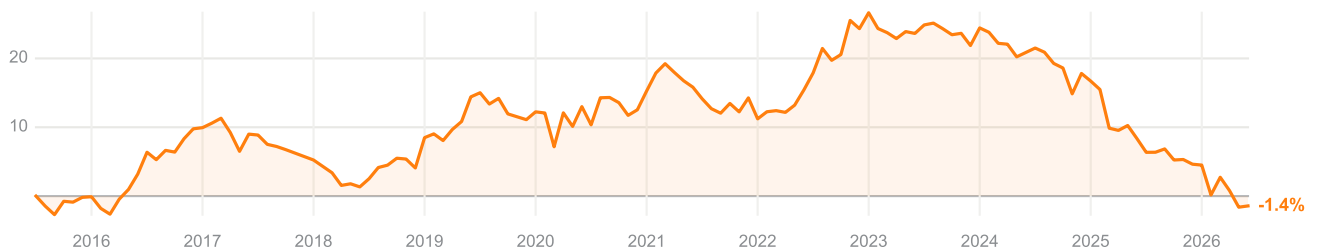
CAGR +0.7%



260 ● 14-Day Relative Strength Index

Price Momentum

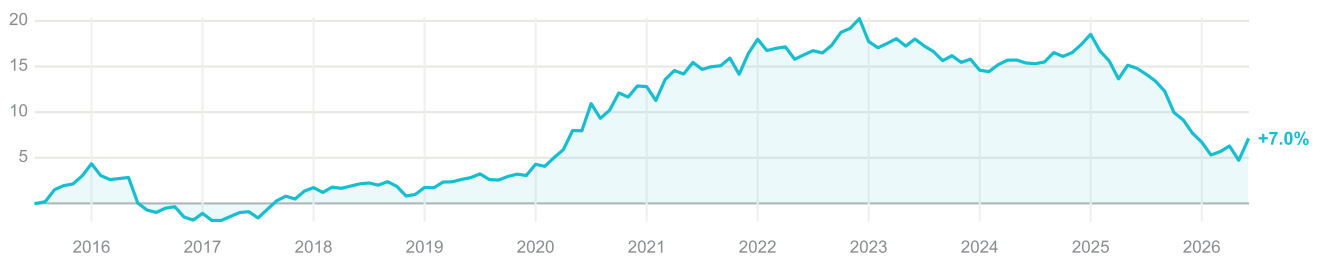
CAGR +0.5%



261 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations

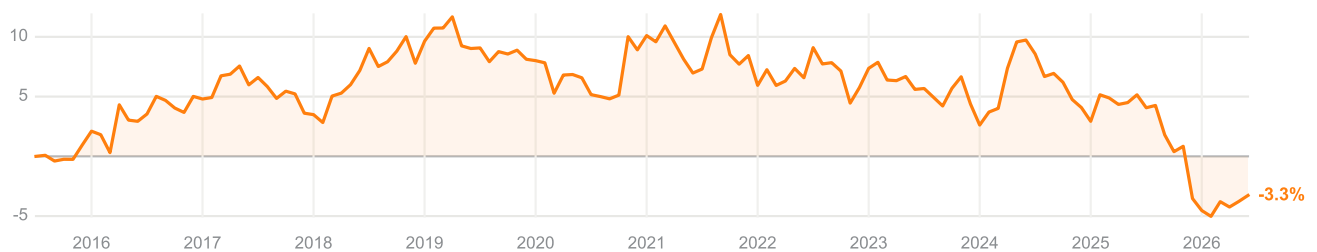
CAGR +0.3%



262 ● 1Y Chg in Share Turnover

Price Momentum

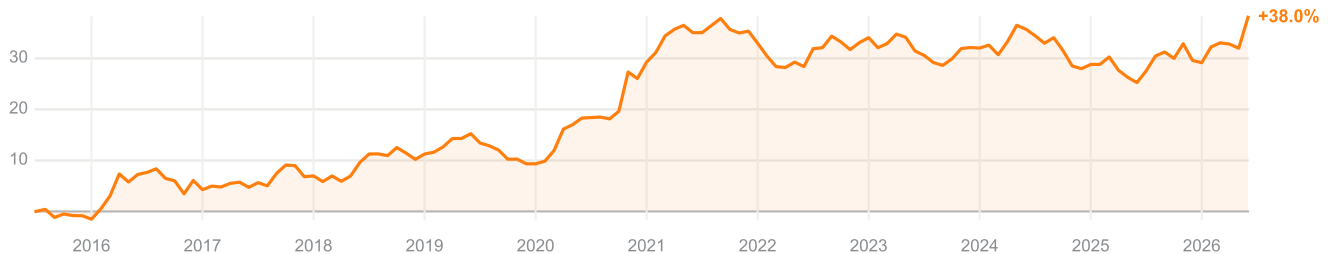
CAGR +0.2%



263 ● 50 Day Volume Signal

Price Momentum

CAGR -0.6%



264 ● Adj 50 Day Volume Signal

Volatility

CAGR -0.6%

